Biochemical and Biophysical Research Communications 828 (2026) 153871

Contents lists available at ScienceDirect

Biochemical and Biophysical Research Communications

journal homepage: www.elsevier.com/locate/ybbrc

Understanding antimicrobial resistance phenotype in Indian clinical isolates through genomic lens

Rupali Aggarwal [a,1], Ankita Das [a,1], Jasleen Kaur [a,1], Anshu Bhardwaj [a,b,*]

[a] Bioinformatics Centre, CSIR-Institute of Microbial Technology, Chandigarh, 160036, India
[b] Academy of Scientific and Innovative Research (AcSIR), Ghaziabad, 201002, India

A B S T R A C T

Antimicrobial Resistance (AMR) poses a global health crisis, with a disproportionately higher burden in developing countries. Yet, the genome-wide exploration of bacterial resistance in high-burden countries like India remains limited due to underrepresentation of high-quality data in publicly available resources. Integrating whole-genome sequencing (WGS) data with standardised phenotypic data, enables characterization of local resistome patterns. In this study, a systematic workflow has been implemented to curate WGS and associated AST data from Indian isolates of critical WHO bacterial priority pathogens. Assembled genomes and curated data were evaluated as per global standards of EUCAST and CLSI. Genomic analyses were performed to identify insertion elements and antibiotic resistance genes (ARGs), correlate presence of ARGs with AST phenotype, and compare the resistomes with global genome datasets for the priority drug-bug combinations. A total of 871 non-redundant AST-linked isolates were curated including *A. baumannii* (n = 119), *E. coli* (n = 305) and *K. pneumoniae* (n = 447) from 75 literature studies, BV-BRC and AST browser. The curated Indian isolates were resistant to many 'Watch' category antibiotics indicating strong reliance on last-resort antibiotics. It was observed that Indian samples are enriched in a small set of resistance determinants like bla_{OXA-23}, bla_{OXA-66} and $bla_{OXA-51-like}$ in *A. baumannii* and $bla_{CTX-M-15}$, bla_{NDM-5}, bla_{TEM-1} in *E. coli* and *K. pneumoniae*, while the global dataset has a broader diversity of ARGs. It is also observed that isolates have more than one resistance mechanism, pointing to AMR being a complex phenotype. ARG co-occurrence networks provide a framework to analyze the complex molecular interactions leading to emergence of AMR. This study highlights the local AMR trends in Indian clinical isolates and provides recommendations for standardized data reporting that are also applicable globally.

1. Introduction

Antimicrobial resistance (AMR) is a natural phenomenon whereby microbes resist the effects of antimicrobials that could previously treat them successfully. This poses a serious concern primarily due to the rapid emergence of resistance to existing antimicrobials. AMR is also referred to as 'a silent pandemic' as it affects millions of people every year but goes unnoticed [1]. The rapid prevalence of resistant bacteria has resulted in high treatment failures, prolonged days of hospitalization, higher mortality and morbidity rates, eventually leading to a global crisis. Recognizing the magnitude of the challenge, WHO has proclaimed AMR as one of the top 10 global public threats. Antibiotic resistance alone can be attributed to around a million human deaths per annum and its yearly toll is expected to rise to ten million lives by 2050 [2]. WHO released Bacterial Priority Pathogens List (BPPL) in 2024 which includes 24 antibiotic-resistance organisms from 15 families categorised into three categories - critical, high and medium - that indicate the global impact and urgency for new interventions (Table S1) [3]. The critical priority group included 3 families of pathogens: carbapenem-resistant *A. baumannii,* third-generation cephalosporin-resistant *Enterobacterales* and carbapenem-resistant *Enterobacterales.* Among these, carbapenem and third-generation cephalosporin resistant *E. coli* and *K. pneumoniae* and carbapenem-resistant *A. baumannii* are highly resistant pathogen-drug combinations and are alone responsible for around 100,000 deaths annually [4]. The rise in AMR cases leads to an increase in health care burden and therefore it is imperative that focused efforts are made to understand the emergence of resistance in these priority pathogens.

Antimicrobial susceptibility testing (AST) has been the gold standard to determine the AMR phenotype, however phenotypic information alone does not offer molecular insights into antimicrobial resistance mechanisms [5]. Moreover, these methods are time-taking and cannot be scaled for wider surveillance and monitoring of AMR. Therefore, in recent years there has been an unprecedented increase in high-throughput DNA sequencing of genomes and metagenomes for bacterial typing, molecular epidemiology investigations, and in-depth pathogenic studies [6]. In addition to identification of new markers to detect microbial infections, whole genome sequencing (WGS) can provide insights into the role of genomic variation in strain evolution and its impact on transmission in real time. In addition, it is also shown that

* Corresponding author. Bioinformatics Centre, CSIR-Institute of Microbial Technology, Chandigarh, 160036, India.
E-mail address: anshu.bhardwaj@csir.res.in (A. Bhardwaj).
[1] All authors equally contributed.

https://doi.org/10.1016/j.bbrc.2026.153871
Received 15 February 2026; Received in revised form 19 April 2026; Accepted 29 April 2026
Available online 5 May 2026
0006-291X/© 2026 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

drug-target interactions are far more complex than previously anticipated, with a single antimicrobial acting via interactions with several proteins in the pathogenic organisms. This also translates towards elucidation of a wide array of mechanisms that can lead to drug resistance. WGS data not only help us to identify the antimicrobial resistance genes, but to also determine insertion elements, virulence factors, mobile genetic elements and also the clonal diversity which gives insights into evolutionary patterns of resistance in the bacterial pathogens.

Studies have evaluated the combined approach of whole genome sequencing (WGS) with analytical tools as a preferred method to reduce time and cost of molecular diagnosis [7]. A large number of these methods have been published and used to infer quantitative relationships between genomic variation and drug resistance phenotypes [8,9]. Moreover, differences in drug-resistant determinants are also observed across different geographies. For example, sequencing and phenotypic characterization of 171 clinical isolates of *Escherichia coli* and *Klebsiella pneumoniae* from Norway and India has been performed using neural networks to predict susceptibility for ampicillin, 3rd generation cephalosporins and carbapenems [10]. This study highlighted several significant challenges from differential presence of AMR genes in data from different sources to lower performance of machine learning models for unseen data as that used in training datasets. The study also highlighted the fact that there is a wide array of resistance mechanisms and therefore, it is recommended to capture genotype-phenotype complexity in a localised context across several pathogen-antibiotic combinations. The challenge posed by AMR is more aggravated in developing countries with high resistance burden combined with cost-constraints and limited diagnostic capacity. Low- and middle-income countries (LMICs) experience up to 90% of the global mortality associated with AMR where there is a high burden of infectious diseases and limited access to advanced treatments, which leads to worsening clinical outcomes, making AMR burden unequally divided among countries [11]. In addition to that, there is a high predominance of data coming from high-income countries even in the publicly available databases [12]. So there are geographical biases in the data as very limited information can be found in the public platforms to study localized AMR mechanisms. There are ongoing initiatives to capture AMR burden (NARS-NET, ICMR report) in India. Also, there has been an increase in the application of multiomics technologies to study Indian clinical isolates [13], however, there is a lack of comprehensive understanding of molecular determinants of AMR in the Indian context.

In this study, systematic curation of published literature as well as databases is performed with specific focus on Indian clinical isolates. The curated datasets comprise of the three critical priority pathogens namely, carbapenem-resistant *A. baumannii,* carbapenem and third-generation cephalosporin resistant *E. coli* and *K. pneumoniae.* The inclusion criteria for clinical isolates included in this study is the availability of AST and genome sequence data. Subsequently, the genomes curated through this process were analyzed for quality metrics as per the EUCAST guidelines [14]. Deep annotation of these curated genomes was performed using standard databases like CARD for the identification of antimicrobial resistance genes (ARGs). Additionally, in-house programs were also developed to elucidate complex resistance mechanisms involving genomic elements other than ARGs. The observations made in this study are presented in light of the quality of genomes and comparison of ARGs in Indian isolates with global trends of ARGs in critical priority pathogens. The distribution of ARGs from the context of resistance in clinical settings across India was observed as per NARS-NET report [15]. It is also observed that isolates have more than one resistance mechanism, pointing to AMR being a complex phenotype. ARG co-occurrence networks are also provided as a framework to analyze the complex molecular interactions leading to emergence of AMR. In the end, recommendations are made to overcome several challenges towards a consolidated understanding of the AMR landscape in the Indian context. To the best of our knowledge, this is the first systematic analysis of the pathogenic strains capturing data from all relevant resources, highlighting scalability of AMR data collection and reporting issues with evidence.

2. Materials and methods

2.1. Data curation

In this study, data was curated for three critical priority pathogens namely, carbapenem-resistant *A. baumannii*, carbapenem and third-generation cephalosporin resistant *E. coli* and *K. pneumoniae*, in Indian context. Data was curated from databases and literature and the steps are described below. Briefly, only those isolates having human as their host, with available WGS data and experimentally reported AST data were included.

2.1.1. Data retrieval & search strategy in databases

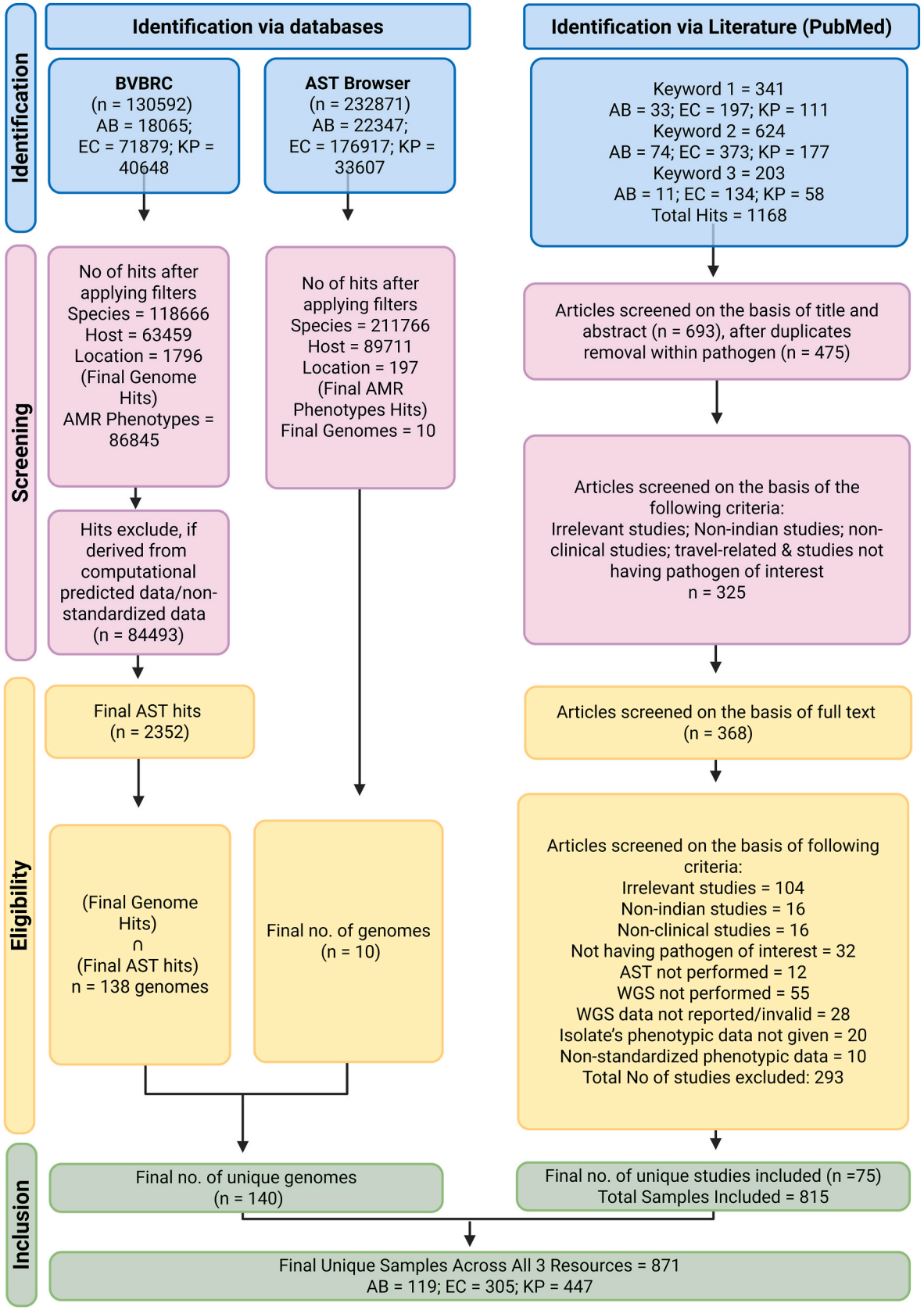
Two databases were referred for curation in this study, namely, BV-BRC [12] and AST Browser (https://www.ncbi.nlm.nih.gov/pathogens/ast). These resources were queried with the species name and subsequently, filters like human host, India as isolation country, availability of WGS and experimentally derived AST were applied in the BV-BRC and AST browser. In BV-BRC, 'Genomes section' was accessed to get genome data and 'Genome ID' was used as an identifier to fetch AST data. Similarly, AST Browser was also searched using the compatible filters as described above. Data from both resources was merged and BioSample ID is used as a common identifier for further analysis (Figs. S1 and S2).

2.1.2. Data retrieval & search strategy for literature curation

A systematic literature curation was conducted in PubMed through keyword-based searches separately for each organism. For *Acinetobacter baumannii*, the following search strings were used: "(((*Acinetobacter baumannii*) OR (*A. baumannii*)) AND ((whole genome sequence) OR (wgs)) AND ((India) or (Indian)))", "(((*Acinetobacter baumannii*) OR (*A. baumannii*)) AND ((genome sequence) AND (resist)) AND ((India) or (Indian)))", "(((*Acinetobacter baumannii*) OR (*A. baumannii*)) AND ((genome sequence) AND (phenotype)) AND ((India) or (Indian)))". Equivalent searches, only replacing the pathogen name, were performed for *E. coli and K. pneumoniae*. The retrieved studies were first screened for their relevance based on their abstract, followed by full-text assessment for extracting WGS accessions and AST information. Hits associated with travel-related studies and non-clinical data were excluded. Studies that provided isolate-level information with publicly accessible WGS data and AST phenotype were included. The PRISMA workflow used for including the studies and samples from the literature and databases is presented in Fig. 1. A standard data structure is designed to capture data with metadata including genome accessions, isolation source, year of collection, phenotypic/AST data and geographic location.

2.2. Genome assembly and quality assessment

Data curated from all three resources were subsequently integrated taking BioSample IDs as the primary identifier and duplicate entries were excluded. For each non-redundant BioSample, NCBI was queried for associated genome assembly accessions and Sequence Read Archive (SRA) run identifiers. The assembled genome FASTA files were directly downloaded from NCBI using the mapped genome accessions. Raw sequencing reads were downloaded for isolates that lacked publicly-available assemblies. The raw reads were checked for quality using FastQC (v0.11.19) [16] and were subjected to quality filtering and pre-processing using fastp (v1.0.1) [17], keeping 20 as Phred score followed by *de novo* assembly using SPAdes (v3.13.1) under default parameters [18]. Average nucleotide identity (ANI) was calculated using fastANI (v1.34) for all three pathogens with default parameters and isolates sharing >95% ANI with respective reference genomes were included for further analysis. Reference based quality assessment was



(caption on next page)

Fig. 1. PRISMA workflow for the selection of studies and samples from the three resources.
The PRISMA workflow illustrates the strategy used to retrieve the WGS data along with available phenotype for Indian clinical isolates using multiple sources: BV-BRC database, AST Browser, and Pubmed for literature reported data. During the Identification phase, records were identified using the keyword search "*Acinetobacter baumannii*" from the BV-BRC database, AST Browser and Pubmed using multiple keywords. During the Screening Phase, for the BV-BRC database: navigate the Keyword search to the Genome section and screen the genome data by applying filters based on Isolation country ("India") and Host ("Human"). To include data from the AMR phenotype section for Indian Clinical isolates only, data files have been mapped, keeping the "Genome ID" column as a common Identifier. For AST Browser, data was screened using aforementioned criteria and literature curation was based on multiple exclusion criteria including travel related studies and non-clinical studies. During eligibility assessment, literature studies were carefully assessed for essential MIC data and other information, including strain accession, MIC values, and phenotypic data. During the final inclusion phase, non-redundant samples from all three resources were included. Duplicate entries were removed at this stage to ensure data non-redundancy. Abbreviations: AB, *A. baumannii*; EC, *E. coli*; KP, *K. pneumoniae*.

done by using QUAST (v5.3.0) [19] for all isolates following the standard guidelines by EUCAST [14]. (Supplementary File 1 & Figs. S3, S4, S5). The complete genomes GCF_009035845.1, GCF_000005845.2, GCF_000240185.1 were used as reference genomes for *A. baumannii* (AB), *E. coli* (EC) and *K. pneumoniae* (KP), respectively.

2.3. Antimicrobial resistance profiling

Antimicrobial susceptibility testing (AST) is a crucial parameter for evaluating the susceptibility of a pathogen to an antibiotic. AST data was extracted from the databases and curated from literature for all the selected isolates. AST data included both Minimal Inhibitory Concentration (MIC) and Disk Diffusion (DD) values. During literature curation, in some studies, only resistant or susceptible phenotypes were provided without AST values (Tables S2, S3,S4 (AST_Phenotype)). These were included in this study provided the study mentions standard guidelines like CLSI/EUCAST while reporting this data (https://clsi.org/shop/standards/m100/) [14]. Where AST data is available, isolates were classified as 'Resistant', 'Susceptible', 'Intermediate' or as 'unclassified strain' (if the reported MIC/DD cannot be classified under any criteria as per CLSI) as per the breakpoints provided in CLSI 2024 using in-house scripts.

2.4. Identification of resistance genes, plasmids and IS elements

The assembled genome sequences (FASTA) were analyzed for antimicrobial resistant genes (ARGs) using the Resistance Gene Identifier (RGI v4.0.1) from the CARD database with default parameters [20]. RGI's perfect and strict hits were considered in this study. For complex resistance mechanisms, insertion elements were also analyzed. Insertion elements were predicted across the samples using ISEScan (v.7.6) [21]. In-house programs have been used to evaluate the role of IS elements in conferring resistance phenotypes as per published reports. Plasmids were reconstructed and annotated using MOB-suite (v3.1.9) for all three pathogens [22]. Plasmids having percentage identity & query coverage >80%, were considered as true-plasmids. In-house scripts were used to identify plasmid-associated ARGs and chromosome-associated ARGs. ARGs having the same contig as the true-plasmid are referred as plasmid-associated ARGs.

2.5. Genotype-phenotype correlation analysis

To evaluate the statistical significance of the associations between the presence of ARGs and the phenotype, Fisher's exact test performed using SciPy library in Python. Genomes were classified into resistant and susceptible groups based on their AST profiles and were integrated with the ARG gene presence-absence matrix, resulting in a 2×2 contingency table for each pair of drug (antibiotic) and ARG. The frequency of each ARG presence in both resistant and susceptible isolates was calculated, along with the corresponding odds ratio (OR) and p-value to evaluate the strength and the direction of the association. A threshold of $P < 0.05$ was applied to identify the statistically significant associations with the resistant phenotype. All the analyses were done by using custom Python scripts.

2.6. Co-occurrence and functional networks of ARGs

Gene co-occurrence networks were constructed to investigate relationships among ARGs for the critical drug-bug combinations listed in WHO BPPL 2024. A binary ARG presence-absence matrix was generated for each isolate, and all pairwise combinations were computed to define co-occurrence relationships. An undirected weighted network was then constructed using Gephi (v0.10.1), in which nodes represented ARGs and edges represented their co-occurrence within the same isolate [23]. Edge weights corresponded to the number of isolates in which each ARG pair co-occurred. Node size was scaled according to ARG prevalence across isolates (i.e., frequency of occurrence). Edge thickness and color intensity were scaled based on edge weights. Nodes were coloured according to their classification into functional groups based on the ARG resistance mechanism.

2.7. ARG comparison between curated and global datasets

The ARGs identified by RGI were compared between the Indian curated genomes and pathogen-specific global datasets. The ARGs of the global datasets were obtained from ARKbase [24]. The Indian isolates were removed from ARKbase's data to prevent overlaps. ARGs were filtered for carbapenem association or cephalosporin association before comparison based on the 'drug class' for each gene in RGI's results. The ARG frequencies were analyzed in both datasets to assess resistance determinants.

3. Results & discussion

3.1. Summary of data curated

As discussed in the Methods section, a systematic workflow was designed and implemented to curate data for understanding AMR landscape in Indian context. Data was curated from databases and literature for the three critical priority pathogens. A total of 138 genomes from BV-BRC and 10 from AST Browser for all three pathogens of interest got shortlisted after preliminary data search from individual resources. During literature curation, 75 studies were shortlisted and included after systematic curation from initial hits of 1168 studies for all three pathogens. After exclusion of duplicates, 693 studies were screened, in which 325 studies were excluded during primary screening on the basis of title and abstract. Later on, 293 studies were excluded during the secondary screening reviewed on the basis of selected inclusion-exclusion criteria after full-text assessment. In the end, 816 genome samples were extracted from the included studies fulfilling the inclusion criteria. For the three pathogens, a total of 872 non-redundant genomes comprising both FASTA and FASTQ (Fig. S1) along with AST data were retrieved with 119, 306 and 447 isolates for *A. baumannii* (Table S2 (AST_MIC, AST_DD)), *E. coli* (Table S3 (AST_MIC, AST_DD)) and *K. pneumoniae* (Table S4 (AST_MIC, AST_DD)), respectively. An *E. coli* sample (GCA_001758055.1) didn't qualify the criteria for species identification as per ANI, hence was excluded. This resulted in 871 genomes which were considered for further downstream analysis. Out of 119 genomes, 29 genomes are complete genomes in *A. baumannii*, 03 out of 305 and 12 out of 447 are complete genomes in *E. coli* and

K. pneumoniae, respectively, as per NCBI. It was observed that very limited data (genomes and AST) on Indian clinical isolates was available from repositories like BV-BRC and AST Browser using their search features. However, a significant number of genomes with AST data were identified from literature on deep curation, reflecting the lack of standardized data reporting on public databases (Fig. S2). Metadata extraction posed an issue as isolation source, geographic location, year of collection along with the performed AST and genome data accessions were either inconsistent or incomplete in many cases. While most metadata on location was extracted using geographic co-ordinates, geographic location for around 10 samples was confirmed through personal communication. Of the 871 samples analyzed, exact geographic location for 73 samples could not be confirmed despite personal communication with the authors. Similarly, the precise isolation source for 77 samples across all three pathogens remained unreported.

3.2. Antimicrobial susceptibility patterns in curated datasets

As the first step towards understanding the patterns of antibiotic resistance in the Indian clinical isolates, they are evaluated from the context of their resistant or susceptible outcomes with respect to various antibiotics. For this study, antibiotics listed in CLSI were considered. In *A. baumannii*, AST in 119 isolates was reported against a set of 21 antibiotics. As seen in Fig. (2A), data on imipenem, trimethoprim/sulfamethoxazole, meropenem, tetracycline, ciprofloxacin, colistin and amikacin in at least 50 isolates is reported. In the majority of these cases, there are more resistant isolates than susceptible ones except in the case of colistin. Among these ciprofloxacin, imipenem and meropenem belong to Watch category and colistin to Reserve category as per AWaRe classification. This represents an alarming situation given that very few new antibiotics are being discovered and emphasis is shifting from the Access to Watch and Reserve category of antibiotics. In the curated data, ~50% samples for *A. baumannii* are reported from blood followed by 15% from respiratory tract associated specimen such as sputum and endotracheal aspirate and remaining 35% of samples are associated with other biological specimens including OSBF (CSF, body fluid), urine, wound & stool samples (Table S2 (Genome_accessions)). This observation seems to be in line with data reported in the National Antimicrobial Resistance Surveillance Network (NARS-NET) report. In isolates reported from blood, around 85% of the samples showed resistance to imipenem and 75% to meropenem which highlights increasing prevalence of carbapenem resistance in *Acinetobacter* species. The AST profile shows increasing resistant phenotype for the first-line and second-line antibiotics and susceptibility to the limited number of third-line or last-resort antibiotics, which reflects alarming insights in the spread of AMR in India.

In case of *E. coli*, there are 305 isolates with AST reported for 43 antibiotics. As seen in Fig. (2B), for the majority of antibiotics resistant isolates are more than susceptible ones except in the case of colistin where no susceptible isolate is reported in this study. Resistance towards colistin which is generally termed as a "last resort" antibiotic is a point of concern. Additionally, being part of third-generation cephalosporins both cefotaxime and cefpodoxime show high levels of resistance but samples still show some sensitivity towards ceftriaxone. All three belong to the Watch category of antibiotics. Approximately, 76% of the samples isolated from blood showed resistance to cefotaxime but around 50% were sensitive for ampicillin (Table S3 (Genome_accessions)). It is also interesting to note the susceptibility towards some antimicrobial agents such as fosfomycin and nitrofurantoin. These antibiotics are found to be appropriate for empiric therapy as they appear to cause minor collateral damage. However, it is also recommended to keep a close monitoring on the emergence of resistance for these antibiotics for sustained use [25]. According to the latest report of NARS-NET, a significant level of resistance has been observed for all specimen types for ciprofloxacin which correlates with the curated data as well. It was also observed that NARS-NET reports that *E. coli* is predominantly found in urine samples, followed by pus aspirate, blood and sterile body fluids. However, in the curated data, majority isolates were sourced from blood samples for *E. coli* (33% in blood followed by ~26% from urine) and 32% from faecal specimens including stool samples, faeces, abdominal infection discharge, rectal and human gut associated samples.

For *K. pneumoniae*, the number of isolates included in this study is the highest with 447 isolates and 44 antibiotics. As can be seen in Fig. (2C), resistant phenotype is seen for almost all antibiotic classes such as carbapenems, fluoroquinolones, aminoglycosides, third and fourth-generation cephalosporins. A limited number of isolates show sensitivity towards few antibiotics such as cefoxitin, ceftazidime/avibactam, meropenem and ertapenem. According to the latest report of NARS-NET, prevalence of *Klebsiella* species has been reported in urine samples followed by pus aspirates, blood and other sterile body fluids. However, in the curated dataset, 40% of *K. pneumoniae* samples were isolated from blood, 16% from urine and 6% from pus and exudates (Table S4 (Genome_accessions)).

In all three pathogens, some strains remain unclassified as the reported MIC/DD values did not fall into any of the defined breakpoints in CLSI 2024 document. This clearly indicates that efforts are needed to harmonise AST protocols and targeted platforms are needed to report AST in alignment with global standards [26]. Another concern is the understanding of 'intermediate' phenotype. EUCAST defines "I" as "Susceptible but increased exposure" but CLSI defines it as "Dose dependent, susceptible" strains which is entirely dependent on dosing of antimicrobial agent and site-specific factors where it concentrates in the body [27]. In all the three pathogens, isolates were present in 'intermediate' and 'unclassified' categories and were not considered in this study.

3.3. Indian resistome landscape and mechanistic insights for critical priority pathogens

In order to understand the resistome landscape of Indian isolates, each isolate was analyzed for the presence of ARGs. The primary analysis focused on the presence of these ARGs in the resistant and susceptible genomes.

3.3.1. Resistome landscape

3.3.1.1. Acinetobacter baumannii. In the case of *Acinetobacter baumannii*, 119 isolates were curated. The resistome analysis of these isolates resulted in 4609 ARG hits with CARD. The total number of unique hits were 133 with perfect and strict criteria (Table S2 (ARGs_AST)). The predicted ARGs belonged to multiple classes of antibiotics, as reported in CARD, ranging from β-lactams, carbapenems, tetracyclines, aminoglycosides and fluoroquinolones. The identified ARGs map to resistance mechanisms, including efflux-pumps, antibiotic target alteration and target inactivation. Of 133, 37 ARGs were annotated for carbapenem resistance. Of these, there were 21 ARGs that were uniquely present in resistant isolates and eight ARGs were commonly found in both resistant and susceptible isolates. On an average, about five to seven ARGs were found per genome, indicating that more than one mechanism may be contributing to resistance phenotype. Given the complex interplay of several factors, multiomics data are needed to provide strong evidence of molecular correlates of resistant phenotypes.

It is observed that all carbapenem-sensitive isolates in this study carried intrinsic $bla_{OXA\text{-}51\text{-}like}$ genes, including $bla_{OXA\text{-}65}$, $bla_{OXA\text{-}106}$, $bla_{OXA\text{-}68}$, $bla_{OXA\text{-}51}$ and $bla_{OXA\text{-}927}$. These findings are in line with the published reports that intrinsic $bla_{OXA\text{-}51\text{-}like}$ β-lactamase genes are not solely responsible for carbapenem resistance [28]. Carbapenem resistance is also mediated by factors such as gene expression levels, gene copy number and the presence of upstream promoter elements [29,30]. To elucidate the role of other factors associated with carbapenem

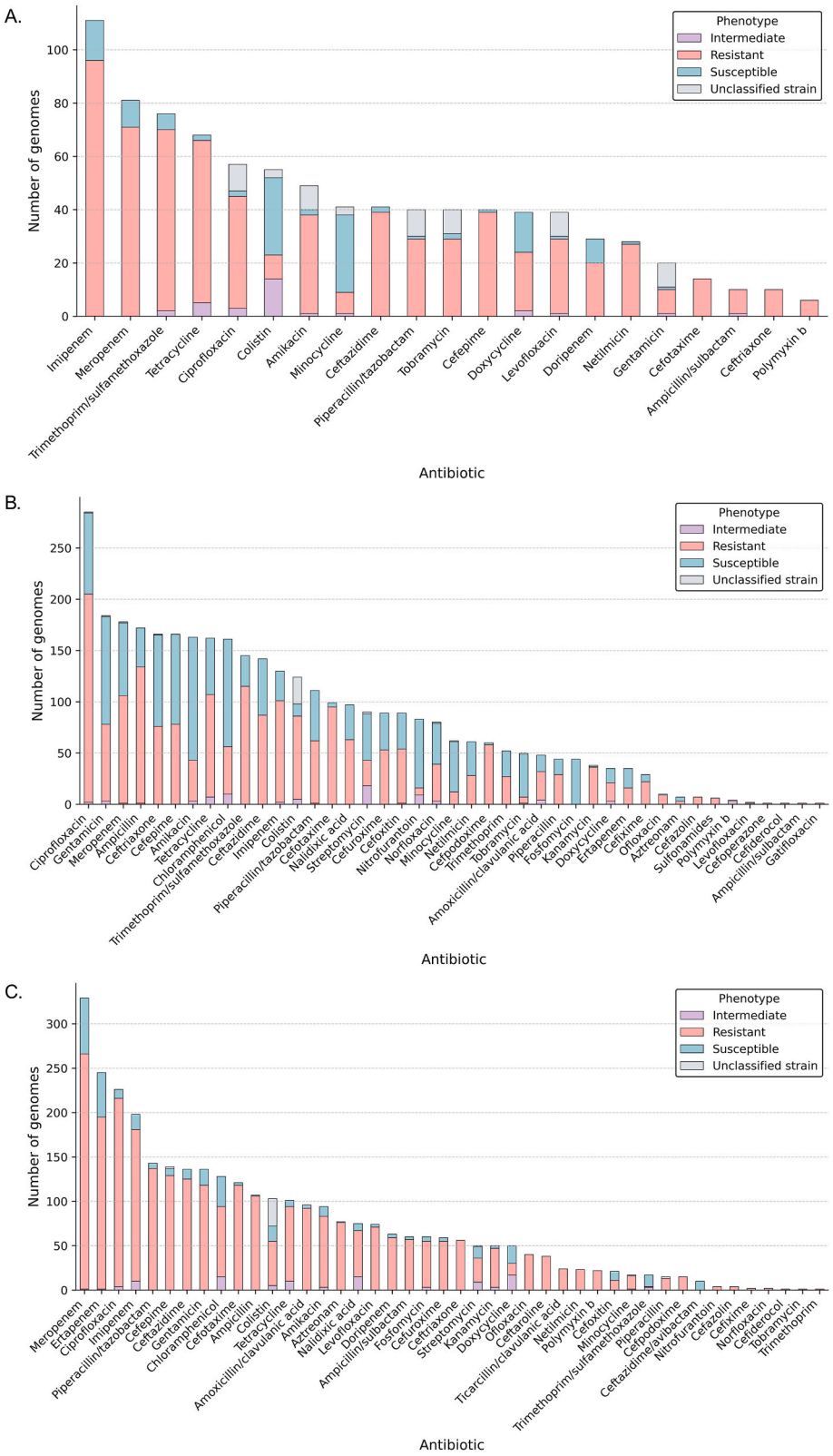


Fig. 2. Antimicrobial Susceptibility Testing (AST) profile across (A) *A. baumannii* (B) *E. coli* and (C) *K. pneumonia* samples
This figure illustrates the number of isolates classified as Resistant (Red), Susceptible (Blue), Intermediate (Purple), Unclassified strains (Grey) for different CLSI recommended antibiotics. This highlights variation in the resistance phenotype for different antibiotic.

resistance and understand why bla_{OXA}-harboring isolates remained phenotypically sensitive, insertion elements were identified for carbapenem-sensitive genomes. Notably, none of the sensitive isolates had insertion elements within 200 bp upstream of $bla_{OXA\text{-}51\text{-}like}$ genes. Previously, many studies have shown that insertion elements such as IS*Aba1* can provide strong promoter sequences, resulting in over-production of bla_{OXA} genes and causing carbapenem resistance. The absence of such upstream insertion elements in sensitive isolates likely explains their inability to express bla_{OXA} genes at levels sufficient to confer resistance [8,31]. Further analysis showed that some ARGs are commonly found in both carbapenem-sensitive and carbapenem-resistant isolates, for e.g., *adeI*, *adeJ*, and *adeK* genes of the adeIJK efflux system, an ancestral efflux pump reported in *Acinetobacter* It is shown that low-level expression of adeIJK plays an important role in the intrinsic resistance of *A. baumannii* to antibiotics like β-lactams, tetracyclines, and fluoroquinolones. In addition to conferring low-level resistance to antibiotics, adeIJK is also reported to contribute to membrane lipid homeostasis in the host to protect against host-generated polyunsaturated fatty acids. This explains why this is an intrinsic factor found in susceptible as well as resistance isolates. In order to gain a better understanding of this efflux pump, one of its known regulators, AdeN, was also explored [32]. AdeN is a transcriptional repressor, which negatively regulates the adeIJK operon, and was detected in all carbapenem-sensitive isolates but only in a few of the resistant isolates. Reports indicate that mutations in regulatory genes do not consistently correlate with clinically significant levels of carbapenem resistance. However, it is proposed that these mutants may offer persistence or tolerance to antibiotics via mechanisms that are not clearly understood. Loss or inactivation of adeN has been previously shown to result in derepression and overexpression of the adeIJK efflux pump, leading to decreased susceptibility to multiple antibiotic classes, including carbapenems [32–34]. Thus, the absence of *adeN* in resistant isolates likely contributes to enhanced efflux activity, whereas its presence in sensitive isolates maintains basal level expression of efflux pumps which is insufficient to exhibit resistance.

In order to explore the association of ARGs with AST, Fisher's exact test was applied. It was observed that 27, 20 and 15 ARGs were found to be significantly associated with resistance phenotype for imipenem, meropenem and doripenem, respectively (Table S2 (Fisher's exact test)). As seen in Fig. 3, $bla_{OXA\text{-}23}$ and $bla_{OXA\text{-}66}$ are the most commonly found variants across all regions. Along with beta-lactamase genes, there was a co-occurrence of cephalosporinase genes like - $bla_{ADC\text{-}30}$, $bla_{ADC\text{-}73}$ genes. Variants like $bla_{ADC\text{-}165}$ and $bla_{ADC\text{-}156}$ were found to be rare and location-specific. The figure not only highlights region-specific ARG patterns, but also identifies the conserved ARG signature genes that were found to be prevalent nationwide. However, given the small sample size, it is imperative to assess these distributions in larger datasets.

The gene presence-absence along with the AST profile as seen in Fig. 4, shows the distribution of significant ARGs ($P < 0.05$; OR > 1, 95% CI) in *A. baumannii* for all three carbapenem drugs. The heatmap for imipenem (Fig. 4 (A)) shows $bla_{OXA\text{-}23}$ is associated with resistance phenotype and is present in 91 out of 96 resistant isolates. There were 94 isolates where either one or both of the genes - $bla_{OXA\text{-}23}$ and $bla_{OXA\text{-}66}$ were present, explaining the resistance phenotype in these samples. Both these genes are globally reported ARGs associated with imipenem resistance. The heatmap shows the presence of various co-occurring resistant determinants as well, including *mphE*, *msrE*, *tetR*, *tet(B)*. While the genes - *tetR* and *tet(B)* are not primary contributors to carbapenem resistance, it has been found in the dataset that both these

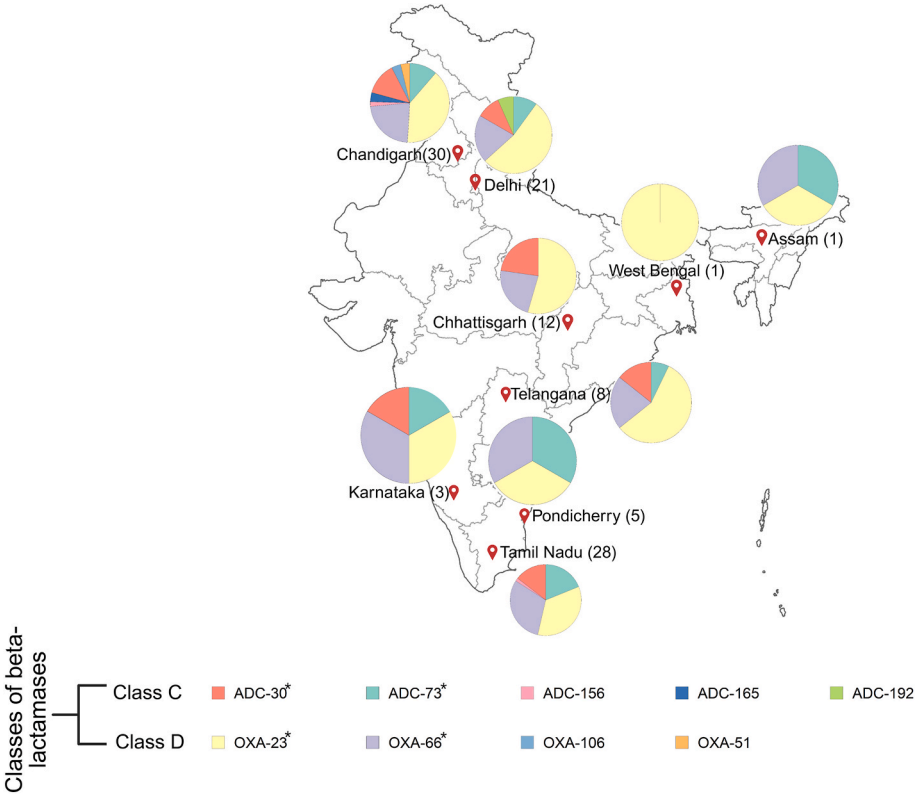
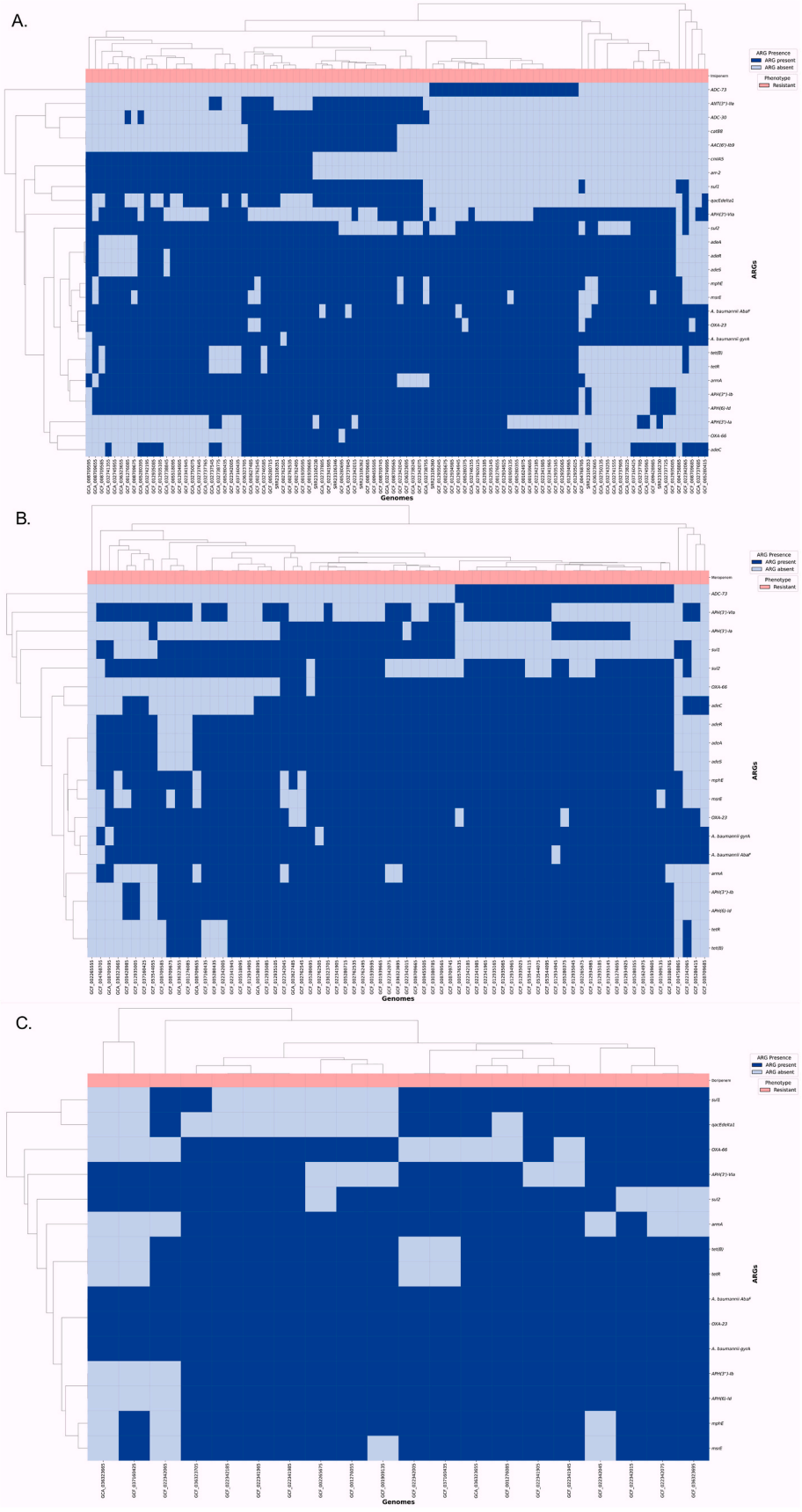


Fig. 3. Geographical distribution of beta-lactamase gene variants in *A. baumannii*
This figure illustrates the geographical distribution of beta-lactamase gene variants in 109 out of 119 *A. baumannii* isolates that were systematically curated in this study. Geographic location for the remaining ten samples were not found. Each pie chart represents the proportion of different beta-lactamase gene variants across different locations. The figure includes only those ARGs that were found to be significant in Fisher's exact test ($P < 0.05$). Genes with asterisks (*) denote that these ARGs are associated with resistance phenotype (OR>1 CI 95%). The numbers in brackets represent the frequency of isolates belonging to the particular geographic location.



(caption on next page)

Fig. 4. Heatmap showing the presence-absence of significant ARGs ($P < 0.05$) and their association with the resistance phenotype (OR > 1, CI 95%) of Imipenem (A), Meropenem (B), Doripenem (C) in *A. baumannii*.
Individual ARGs are represented by the rows, and genomes are represented by columns. The heatmap shows gene presence (dark blue) and absence (light blue). Genomes are hierarchically clustered based on ARG profiles using distance-based clustering (dendrogram shown on the top). The red bar on the top shows the resistant phenotype of the genomes The heatmap contains only those ARGs that showed a statistically significant association with the resistance phenotype of imipenem, meropenem and doripenem ($P < 0.05$, Fisher's exact test).

genes are statistically significant for tetracycline resistance as well as carbapenem resistance, which leads to a challenge in differentiating drug-specific markers. On further investigation, it was found all doxycycline resistant samples were also carbapenem resistant, showing co-selection of both carbapenem and tetracycline resistant genes in the isolates. Since, all the doxycycline resistant samples are also carbapenem resistant, but not all carbapenem resistant samples are doxycycline resistant, doxycycline may be used in treatment of carbapenem resistant isolates, as has been reported earlier [35].

Of the total 96 resistant isolates, two samples lacked both $bla_{OXA\text{-}23}$ and $bla_{OXA\text{-}66}$ genes highlighting the fact there might be other mechanisms responsible for imipenem resistance in these isolates. While one of the isolates harbors *blaOXA* variants like $bla_{OXA\text{-}94}$ (belonging to intrinsic OXA 51-like family) and $bla_{OXA\text{-}420}$ (belonging to OXA-58 like family), the other isolate harbors $bla_{OXA\text{-}144}$ gene. These *blaOXA* genes are not found to be statistically significant for resistance and are also found in a few isolates in the Indian dataset. It is to be noted that both samples are also resistant to most cephalosporin drugs and they harbor several ADC genes including $bla_{ADC\text{-}176}$, $bla_{ADC\text{-}76}$ and $bla_{ADC\text{-}80}$. It may be hypothesized that in these samples cephalosporinase might also be responsible for carbapenem resistance, if overexpressed, and needs to be further investigated. Similarly, among meropenem resistant samples (71 samples) (Fig. 4 (B)), all except for three isolates harboured either $bla_{OXA\text{-}23}$ or $bla_{OXA\text{-}66}$ genes, explaining the resistance phenotype in the samples. Of these three isolates, two isolates were same as discussed above and, the remaining isolate has bla_{OXA98}, which is an intrinsic class D OXA-51-like-lactamase and may contribute to resistance to meropenem. All the doripenem resistant samples [20] (Fig. 4 (C)), harboured $bla_{OXA\text{-}23}$ and $bla_{OXA\text{-}66}$ explaining their resistant phenotype.

In the susceptible samples, a total of six significant ARGs ($bla_{OXA\text{-}106}$, $bla_{OXA\text{-}51}$, $bla_{ADC\text{-}156}$, $bla_{ADC\text{-}165}$, $bla_{ADC\text{-}192}$, *adeN*) (Figs. S6 and S7) were found to be negatively associated with resistance for the three carbapenem drugs, which highlights discordance in the genotype-phenotype correlation. Despite the presence of beta-lactamase genes and cephalosporinase genes, these samples show phenotypic susceptibility. This observed discordance in the data shows that the presence of these genes is not sufficient to cause resistance in the isolates. Also, there might be a possibility that these samples showing susceptibility may be potential resistant isolates and needs to be followed up over time for deeper understanding of emergence of resistance.

According to the annual AMR report by ICMR, the co-occurrence of $bla_{OXA\text{-}23}$ and bla_{NDM} genes in the samples is reported as a concerning combination of variants, as this pattern causes resistance to almost all beta-lactam drugs, leading the evolution of superbugs in *A. baumannii* population. When investigated in the study dataset, it is observed that out of 96 resistant samples, 13 isolates harboured this combination of ARGs. Also, it is important to note that these ARGs are known to be plasmid mediated as also observed in this study. Additionally, co-occurrence of $bla_{OXA\text{-}23}$ in the chromosome and plasmid in *A. baumannii* has been shown to improve bacterial survival [36]. In this study, occurrence of $bla_{OXA\text{-}23}$ in the chromosome and plasmid in *A. baumannii* was found in only one sample (Table S2 (Plasmid)).

3.3.1.2. Escherichia coli. In the case of *Escherichia coli*, 305 genomes were curated. The resistome profile of the genomes resulted in a total hit of 19565 ARGs, with 178 unique hits (Table S3 (ARGs_AST)). The ARGs were found to belong to multiple drug classes including aminoglycosides, fluoroquinolones, carbapenems, and cephalosporins. Similar to *A. baumannii*, the identified resistance mechanisms include efflux-pumps, antibiotic target alteration and target inactivation. For carbapenems and cephalosporins, 52 unique ARGs were predicted in *E. coli* isolates. Of these, 13 ARGs were specific to resistant isolates and 34 were common to both resistant and susceptible isolates.

Most of the curated genomes have genes *acrB* and *tolC* along with the regulatory gene *marA*, which shows how these genes from the efflux pump systems have a basic role in resistance mechanism and can lead to reduced susceptibility to multiple drug classes. Among the extended-spectrum beta lactamases (ESBLs), the most prevalent variant was $bla_{CTX\text{-}M\text{-}15}$, followed by $bla_{CTX\text{-}M\text{-}27}$ and $bla_{CTX\text{-}M\text{-}14}$. CTX-M enzymes are by far the most commonly found ESBLs in *E. coli* which is also seen in this study as $bla_{CTX\text{-}M\text{-}15}$ is present in 50% of the samples. Isolates carrying this gene were seen to be resistant to both ceftriaxone and cefotaxime, except for a few isolates. The $bla_{CTX\text{-}M\text{-}15}$ gene which is a part of the class-A extended-spectrum beta lactamases along with the insertion element IS*Ecp1* (IS*I380* family) located downstream has been reported to confer resistance to third-generation cephalosporins including ceftriaxone and cefotaxime [37]. However, IS*Ecp1* insertion element within close proximity (up to 200bp) of $bla_{CTX\text{-}M\text{-}15}$ gene could not be identified, suggesting that the mechanism of $bla_{CTX\text{-}M\text{-}15}$ with IS*Ecp1* is not the sole mechanism responsible for the resistance phenotype [38]. As suggested in the literature, many other factors like the presence of secretion systems are also responsible for the overexpression of the CTX-M enzymes [9]. In the curated dataset, T2SS was observed in all isolates and may be associated with the resistance phenotype.

It was observed that 13, 29, 24, 5 and 31 ARGs were found to be significantly associated with resistance phenotype ($P < 0.05$ and OR > 1, 95% CI) for imipenem, meropenem, ertapenem, cefotaxime and ceftriaxone, respectively (Table S3 (Fisher's_Exact_Test)) (Figs. S8, S9, S10, S11).

As seen in Fig. 5, $bla_{TEM\text{-}1}$, $bla_{CTX\text{-}M\text{-}15}$, $bla_{NDM\text{-}5}$, were the most commonly found variants across all regions. Variants like $bla_{EC\text{-}18}$, $bla_{EC\text{-}19}$ were found to be rare and location-specific.

The heatmap represented in Fig. 6 shows gene presence-absence along with AST profile of the cephalosporin drugs in *E. coli*. Among cephalosporinases, $bla_{CTX\text{-}M\text{-}15}$ was found to be most prevalent. Out of 95 cefotaxime resistant samples, 67 samples harbour this gene, explaining resistance in all of these samples, as this variant is most commonly reported in cephalosporin resistance (Fig. 6 (A)). Other commonly found genes were $bla_{OXA\text{-}1}$ and *BRP* (MBL) and these were found positively associated with cephalosporin resistance in *E. coli*. However, in two samples neither of these variants and interestingly, none of the bla_{CTXM} variants, which are known as hydrolyzing enzymes of cephalosporin drugs like cefotaxime and ceftriaxone, were found. While, these samples have the commonly found efflux systems like AcrAB-TolC, along with their regulatory genes *marA*, *soxR* and *soxS*, they lack any bla_{OXA} variants which primarily are not responsible for cephalosporin resistance, but when overexpressed can cause resistance to these drugs [39,40].

Similarly, in ceftriaxone-resistant samples (Fig. 6 (B)), $bla_{CTX\text{-}M\text{-}15}$ was found to be highly associated with the resistance phenotype, explaining resistance in 35 samples out of total 76 resistant isolates. There were other beta-lactamases genes that were also found to be prevalent in the dataset, including $bla_{CMY\text{-}42}$, $bla_{NDM\text{-}5}$ and $bla_{OXA\text{-}1}$. Notably, *BRP* (MBL) which is not primarily responsible for ceftriaxone resistance in *E. coli*, was also found to be highly associated with ceftriaxone resistance in the dataset. Certain genes like $bla_{CMY\text{-}132}$ and $bla_{NDM\text{-}7}$ showed dispersed distribution in the resistant isolates.

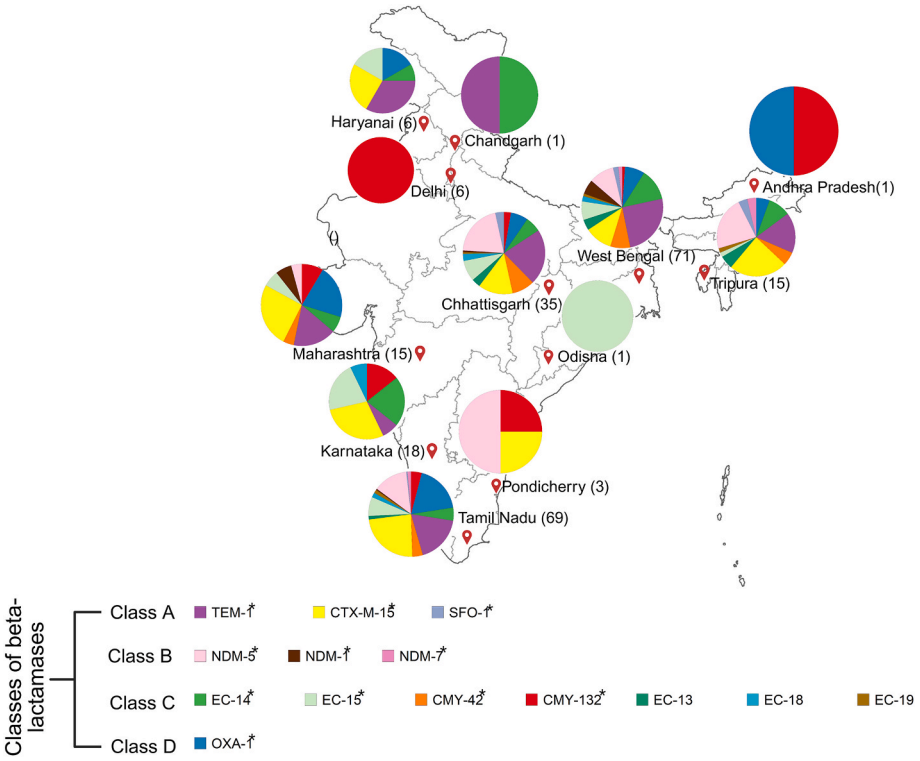


Fig. 5. Geographical distribution of beta-lactamase gene variants in *E. coli*.
This figure illustrates the graphical distribution of beta-lactamase gene variants in 241 out of 305 *E. coli* isolates that were systematically curated for the study. The figure includes only those ARGs that were found to be significant ($P < 0.05$, Fisher's exact test). Isolates with the significant ARGs and known geographic location are shown here. Each pie chart here represents the proportion of different beta-lactamase gene variants found in each geographic location of India. Genes with asterisks (*) denote those ARGs that are found to be associated with resistance phenotype (OR>1, 95% CI). The numbers in brackets represent the frequency of isolates belonging to that particular geographic location.

In the case of carbapenem resistance in *E. coli* Figure (7A, B, C), $bla_{CTX\text{-}M\text{-}15}$ was found to be highly associated with the resistance phenotype of both meropenem and imipenem, but was not identified in any of the ertapenem resistant samples. The genes $bla_{TEM\text{-}1}$ and $bla_{NDM\text{-}5}$ were most commonly present across all three carbapenem drugs, explaining resistance in 93 samples out of total 116 carbapenem-resistant isolates. Some variants were found to be associated with carbapenem resistance in the dataset like - $bla_{NDM\text{-}1}$, $bla_{NDM\text{-}7}$, $bla_{CMY\text{-}42}$, showing that there is a diverse distribution of beta-lactamase genes that contribute to resistance to carbapenem in the Indian dataset. In the resistome analyses, it was observed that $bla_{CTX\text{-}M\text{-}15}$, $bla_{TEM\text{-}1}$, $bla_{OXA\text{-}1}$, and $bla_{NDM\text{-}1}$ were most prevalently found in the Indian dataset, which is in line with the findings reported in the ICMR report as well as in a recent surveillance study done on Indian genomes [41,42]. The report states that TEM, NDM, OXA, VIM, IMP, and CTX-M class of genes are the most prevalent and critical to understand AMR evolution in *E. coli*. These genes are reported as predominantly associated with plasmid-mediated resistance in clinical enterobacteriaceae [43]. Among these, bla_{TEM}, $bla_{NDM\text{-}1}$, $bla_{NDM\text{-}5}$ and $bla_{OXA\text{-}1}$ were found to be associated with plasmid-mediated resistance, while $bla_{NDM\text{-}4}$ was found to be chromosome-associated only in the dataset (Table S3 (Plasmid)).

3.3.1.3. Klebsiella pneumoniae. In the case of *K. pneumoniae*, 447 curated genomes were analyzed. The resistome profile for the *K. penumoniae* genomes resulted in a total hit of 20,061 ARGs, with 157 unique hits. Fifty-three ARGs were uniquely present for carbapenems and cephalosporins. (Table S4 (ARGs_AST))

As seen in Fig. 8, two Class A beta-lactamase genes - $bla_{CTX\text{-}M\text{-}15}$, $bla_{TEM\text{-}1}$ were present in ~67% of the isolates across all geographic locations of India. Along with that $bla_{NDM\text{-}5}$ and $bla_{OXA\text{-}232}$ were also found widely distributed in most of the geographic locations. Despite the variation in sample size, genes that were found to be significantly associated with resistant phenotypes ($P < 0.05$, OR > 1, 95% CI) like $bla_{CTX\text{-}M\text{-}15}$, $bla_{TEM\text{-}1}$, $bla_{NDM\text{-}5}$, $bla_{OXA\text{-}232}$ were found in most of the regions indicating that these are the main drivers of the resistance. The bla_{SHV} variants like $bla_{SHV\text{-}75}$, $bla_{SHV\text{-}27}$ and $bla_{SHV\text{-}31}$ were found to be restricted to specific-locations only.

The heatmap represented in Fig. 9 shows gene presence-absence along with the AST profile of the carbapenem drugs in *K. pneumoniae*. The Fisher's exact test identified 16, 39 and 40 significant ARGs that are associated with resistance phenotype in imipenem, meropenem and ertapenem, respectively (Table S4 (Fisher's_Exact_Test)) (Figs. S12, S13, S14). As seen in Figure (9A), the most common ARG for imipenem resistance was *kpnE*, which is an efflux pump and is present in all the 169 resistant isolates. The gene *pmrF*, which is known for polymyxin resistance, was present in all the isolates, except for three. The other predominantly found genes were $bla_{TEM\text{-}1}$, $_{sul1}$, $bla_{OXA\text{-}232}$ and $bla_{SHV\text{-}28}$. For ertapenem, the genes $bla_{CXT\text{-}M\text{-}15}$, $bla_{OXA\text{-}232}$ and $bla_{TEM\text{-}1}$ were the most prevalent Figure (9B). Samples not harbouring these variants have either NDM or other OXA variants like $bla_{OXA\text{-}1}$ or $bla_{OXA\text{-}181}$, explaining their resistance phenotype to carbapenems. Similarly, for meropenem, $bla_{CTX\text{-}M\text{-}15}$ and $bla_{TEM\text{-}1}$ were the most prevalent ARGs, followed by $bla_{OXA\text{-}181}$, $bla_{OXA\text{-}1}$ and *sul2* genes Figure (9C).

For cephalosporin drugs, there was very limited data for sensitive samples. So while Fisher's test identified significantly associated ARGs for cefotaxime, cefuroxime drugs, no significant ARGs were identified for ceftriaxone (Figs. S15 and S16).

As seen in the heatmap Fig. 10, *pmfr*, *vanG*, *sul1*, *ompA* and *kpnE* were found to be highly associated with both cefuroxime and cefotaxime resistance. All the cephalosporin resistant samples harbour either of these variants, explaining the resistant phenotype in these samples.

As reported by ICMR, bla_{TEM}, $bla_{OXA\text{-}48}$, bla_{SHV}, bla_{NDM} were mostly

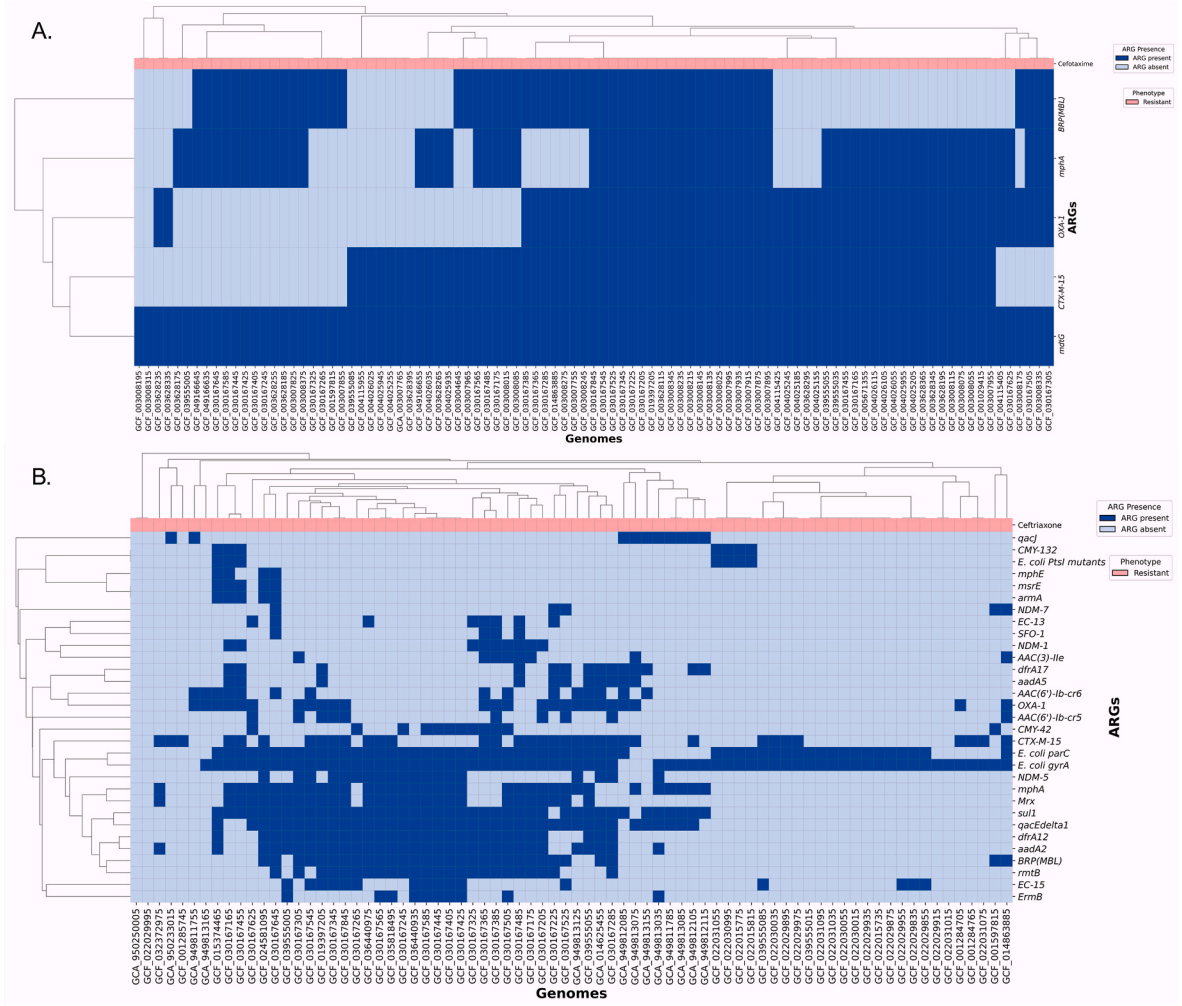


Fig. 6. Heatmap showing the presence-absence of significant ARGs ($P < 0.05$) and their association with the resistance phenotype (OR > 1, CI 95%) of Cefotaxime (A), Ceftriaxone (B) in *E. coli*.
Individual ARGs are represented by the rows, and genomes are represented by columns. The heatmap shows gene presence (dark blue) and absence (light blue). Genomes are hierarchically clustered based on ARG profiles using distance-based clustering (dendrogram shown on the top). The red bar on the top shows the resistant phenotype of the genomes. The heatmap contains only those ARGs that showed a statistically significant association with the resistance phenotype of cefotaxime and ceftriaxone ($P < 0.05$, Fisher's exact test).

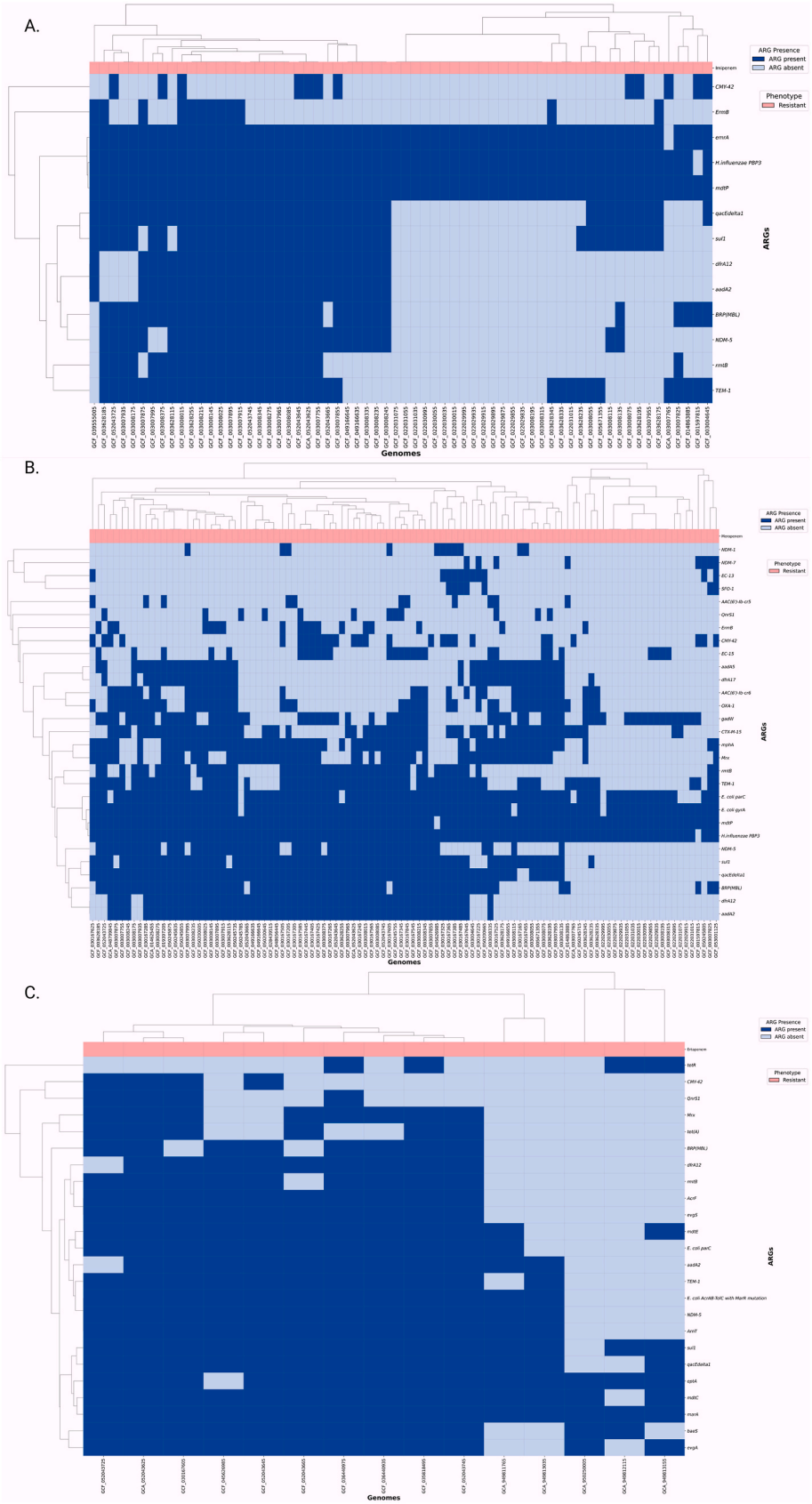
found beta-lactamase resistance genes. According to the reported literature, these genes are predominantly found associated with plasmid mediated resistance mechanisms [44]. Among these, bla_{TEM}, bla_{SHV}, bla_{NDM} were found to be associated with plasmid-mediated resistance in the dataset (Table S4 (Plasmid)).

To the best of our knowledge, this is the first study focusing on systematic curation and analysis of WHO priority pathogens in Indian context.

3.3.2. *ARG co-occurrence and functional networks*

Resistance gene co-occurrence networks, in which ARGs are represented as nodes connected based on their frequency of co-occurrence within an isolate, were constructed *for A. baumannii, E. coli* and *K. pneumoniae* (Tables S2, S3, S4 (Networks)). Across all three organisms, a clear distinction between a conserved core set of ARGs and a heterogenous accessory set can be observed (Fig. 11 (A), (B), (C)). A set of ARGs are uniformly present across most isolates, particularly those involved in multidrug efflux systems and target-modifying enzymes. The most frequent and conserved ARGs, bigger nodes, exhibit high co-occurrence with the rest of the genes and take central positions in the network. On the other hand, a large number of ARGs are sparsely distributed across isolates and take up peripheral positions around the conserved genes in the network. This feature is consistent across all three organisms, suggesting that antimicrobial resistance follows a shared systems-level architecture despite differences in gene composition. Functional characterization of the nodes has helped visualize the interactions between different resistance genes. This representation provides a framework to understand how multiple resistance mechanisms co-occur within individual genomes rather than acting in isolation. The resulting network indicates a clear distinction between highly conserved ARGs, particularly those involved in multidrug efflux systems (such as *adeIJK*, *adeABC*, *abeS*, etc.) and target-modifying enzymes (such as *gyrA* and *parC*), and sparsely distributed ones including several β-lactamase variants. The network also shows high frequency of co-occurrence between different resistance classes, highlighting the combinatorial nature of AMR mechanisms. The conserved and intrinsic mechanisms give a foundational level of resistance and accessory ARGs additionally contribute to it. These associations support that resistant phenotypes arise from coordinated interactions between multiple mechanisms.

Multidrug efflux systems represent a highly conserved component of the resistomes, as reflected by their high prevalence within the co-occurrence network. For *A. baumannii*, RND efflux pumps, particularly *adeABC*, *adeIJK*, and *adeFGH*, are the most clinically relevant. These all



(caption on next page)

Fig. 7. Heatmap showing the presence-absence of significant ARGs ($P < 0.05$) and their association with the resistance phenotype (OR > 1, CI 95%) of Imipenem (A), Meropenem (B), Ertapenem (C) in *E. coli*.
Individual ARGs are represented by the rows, and genomes are represented by columns. The heatmap shows gene presence (dark blue) and absence (light blue). Genomes are hierarchically clustered based on ARG profiles using distance-based clustering (dendrogram shown on the top). The red bar on the top shows the resistant phenotype of the genomes. The heatmap contains only those ARGs that showed a statistically significant association with the resistance phenotype of imipenem, meropenem, ertapenem ($P < 0.05$, Fisher's exact test).

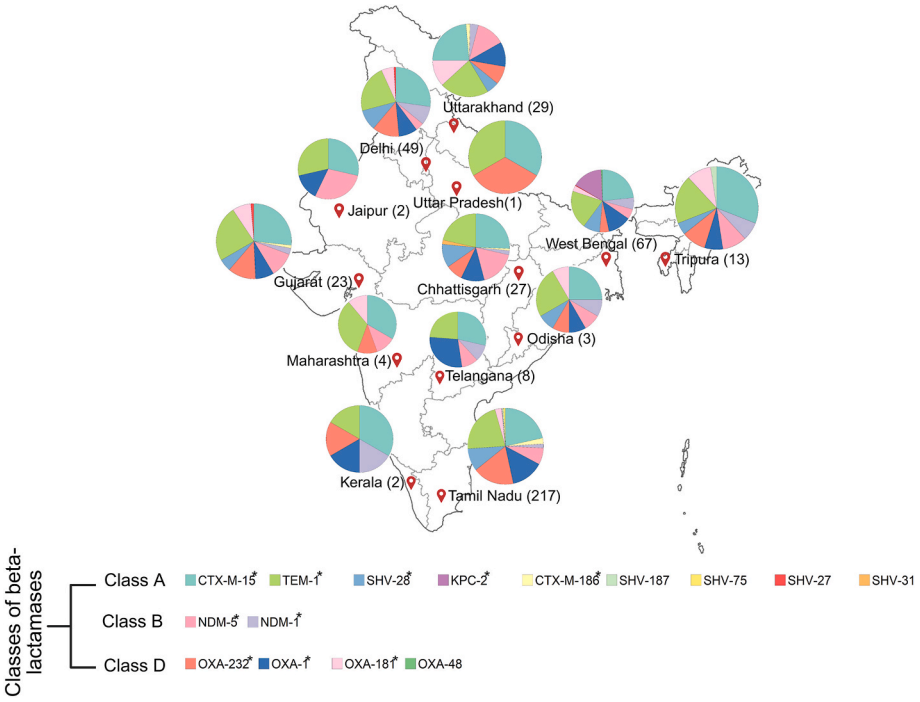


Fig. 8. Geographical distribution of beta-lactamase gene variants in *K. pneumoniae*.
This figure illustrates the graphical distribution of beta-lactamase gene variants in 445 out of 447 in *K. pneumoniae* isolates that were systematically curated for the study. The figure includes only those ARGs that were found to be significant ($P < 0.05$, Fisher's exact test). Each pie chart here represents the proportion of different beta-lactamase gene variants found in each geographic location of India. Genes with asterisks (*) denote those ARGs that are found to be associated with resistance phenotype (OR>1 95% CI). The numbers in the brackets represent the frequency of isolates belonging to that particular geographic location. No significant ARGs were identified in two samples that originated from Karnataka.

take central positions in the network. Similarly, the MFS EmrAB-TolC system in the *E. coli* network, the *oqxAB* efflux genes in the *K. pneumoniae* network, and other genes of the RND efflux system (*acrAB*), ABC transporter complexes (*kpnEFGH*) and MFS efflux systems (mdfA and mdt family) in both *E. coli* and *K. pneumoniae*, are present in the conserved central sets. Along with the efflux genes, their associated regulators, such as adeN for adeIJK system, and global regulators like *marA*, *CRP*, and *rsmA* are also present in high abundance across the three organisms.

β-lactam resistance in *A. baumannii* is predominantly mediated by class D β-lactamases (OXA-type) which exhibit a wide variety of variants across isolates, possibly reflecting evolutionary divergence and mobility events. Different OXA families vary in substrate specificity, catalytic efficiency, and regulatory context. While these enzymes all act by hydrolyzing β-lactam antibiotics, there are key distinctions among them. While $bla_{OXA\text{-}23}$ is the most prevalent, the intrinsic $bla_{OXA\text{-}51\text{-}like}$ genes have the highest diversity across samples. In *E. coli*, $bla_{TEM\text{-}1}$ and $bla_{CTX\text{-}M\text{-}15}$ are the most prevalent β-lactamases, indicating high ESBL-mediated resistance. *E. coli* network also highlights the carbapenemase bla_{NDM} variants, particularly $bla_{NDM\text{-}5}$, and the class D β-lactamase $bla_{OXA\text{-}1}$. Unlike the other two organisms, where β-lactamases are mostly near the network's periphery, a substantial proportion of *K. pneumoniae* isolates harbor them in high-frequency. $bla_{CTX\text{-}M\text{-}15}$ dominates the ESBL group, supported by blaTEM and blaSHV variants and *K. pneumoniae* isolates also harbour carbapenemases, including *blaNDM* variants, $bla_{KPC\text{-}2}$ and $bla_{OXA\text{-}48\text{-}like}$ enzymes such as $bla_{OXA\text{-}181}$ and $bla_{OXA\text{-}232}$. These belong to different classes of β-lactamases, and their co-occurrence in the same isolate broadens its resistance capabilities.

While applying statistical analyses is valuable, it is influenced by gene prevalence in the dataset and low frequency variants might not get identified as significant contributors to resistance despite their functional importance, both individually and in combination with other ARGs. Gene presence-absence analyses across species revealed useful information on the components of the resistome, but the network-based analysis has provided a systems perspective into how these components interact within individual isolates. The co-occurrence patterns and the underlying interactions between different resistance genes in the study highlight that understanding and accurately interpreting resistance phenotypes require systems-level perspective integrating omics-based approaches.

3.4. Comparison with global ARG dataset

3.4.1. A. baumannii

A comparison was made between carbapenem resistance genes found in this study (119 *A. baumannii* isolates) with a non-overlapping global dataset (635 *A. baumannii* genomes) sourced from Antimicrobial Resistance Knowledgebase (ARKbase). The global dataset showed greater resistome diversity with 75 unique carbapenem-associated ARGs, whereas in Indian isolates 37 unique carbapenem-associated

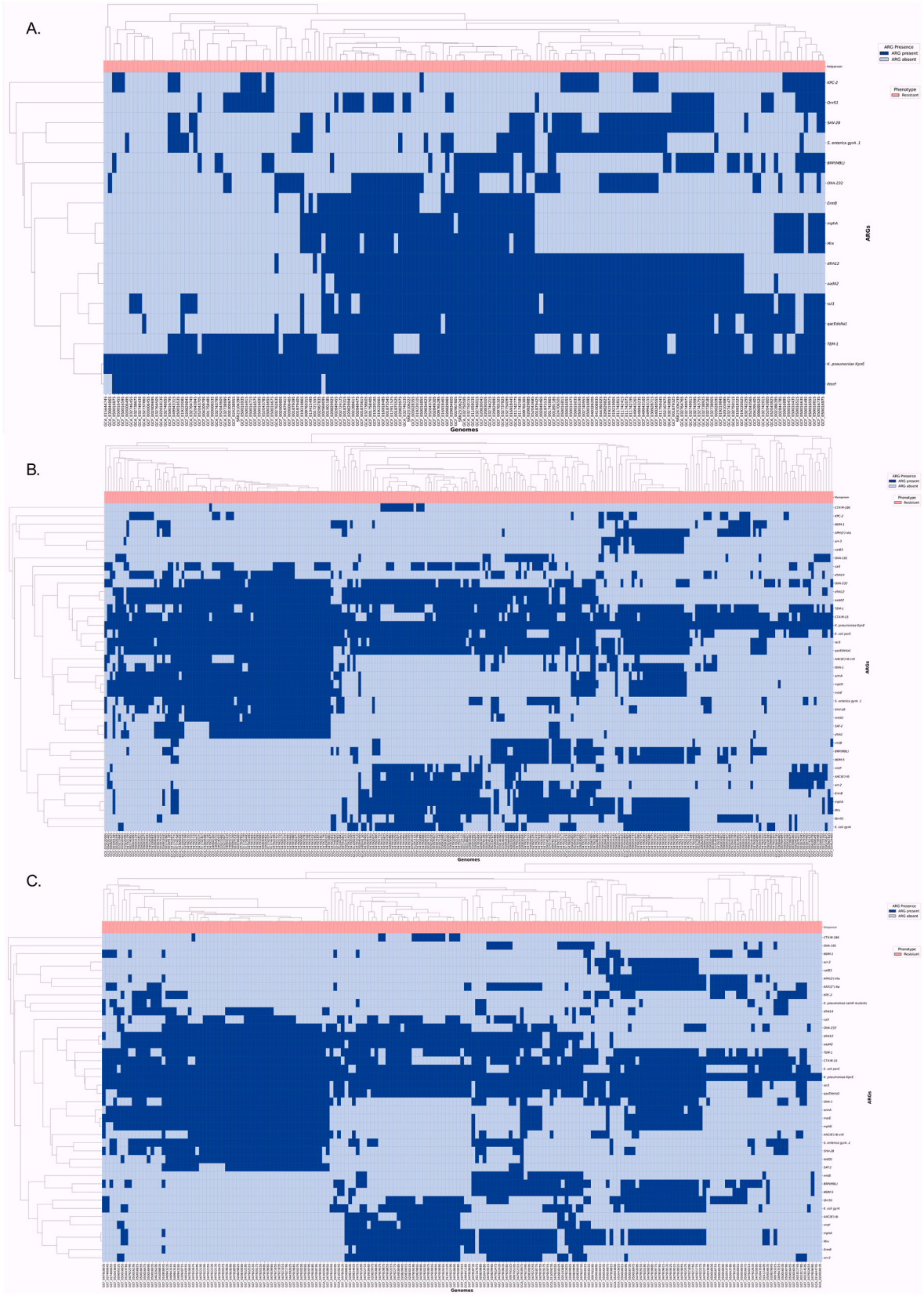


Fig. 9. Heatmap showing the presence-absence of significant ARGs ($P < 0.05$) and their association with the resistance phenotype (OR > 1, CI 95%) of Imipenem(A), Meropenem (B), Ertapenem (C) in *K. pneumoniae*.
Individual ARGs are represented by the rows, and genomes are represented by columns. The heatmap shows gene presence (dark blue) and absence (light blue). Genomes are hierarchically clustered based on ARG profiles using distance-based clustering (dendrogram shown on the top). The red bar on the top shows the resistant phenotype of the genomes. The heatmap contains only those ARGs that showed a statistically significant association with the resistance phenotype of imipenem, meropenem, ertapenem ($P < 0.05$, Fisher's exact test).

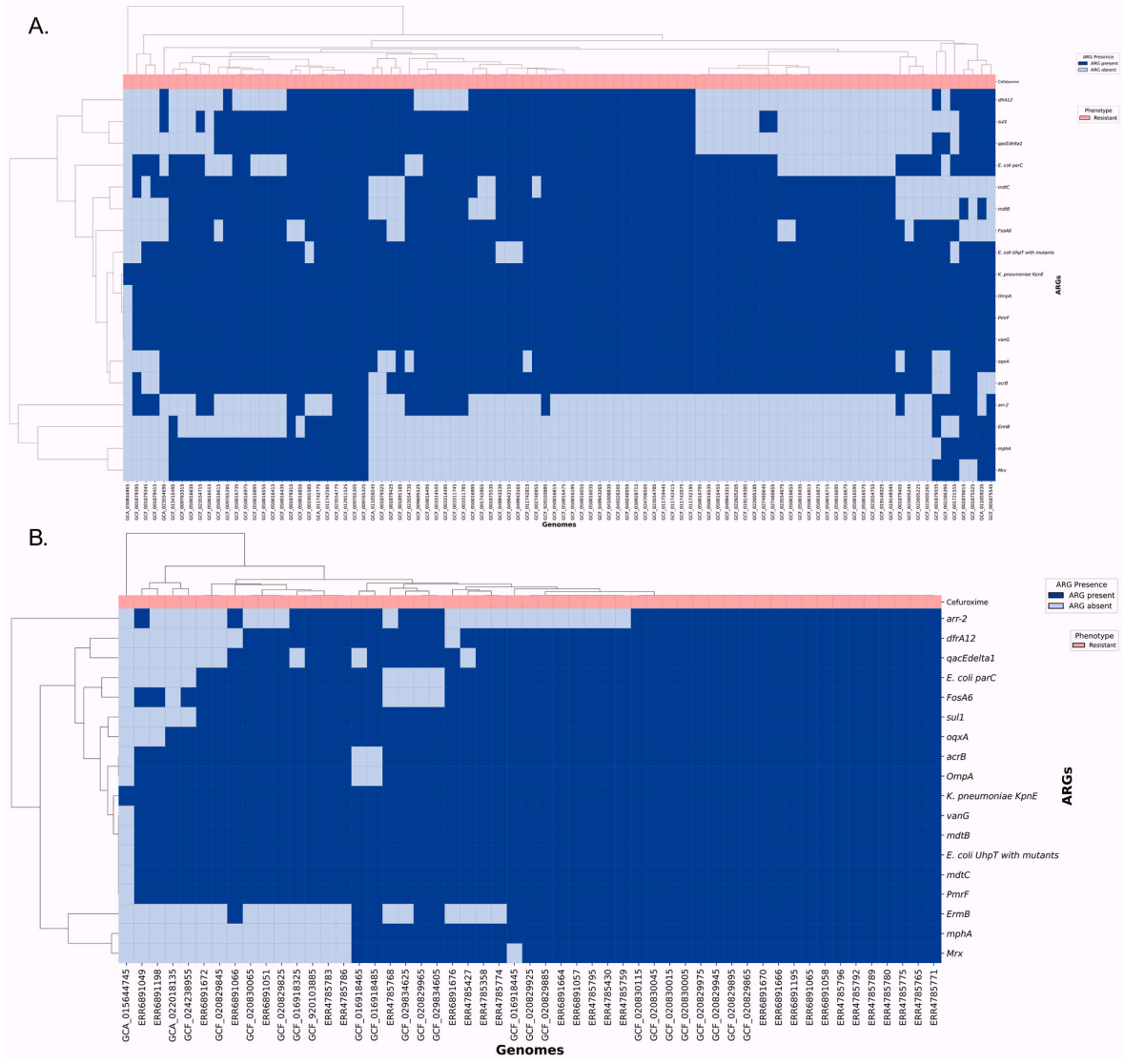
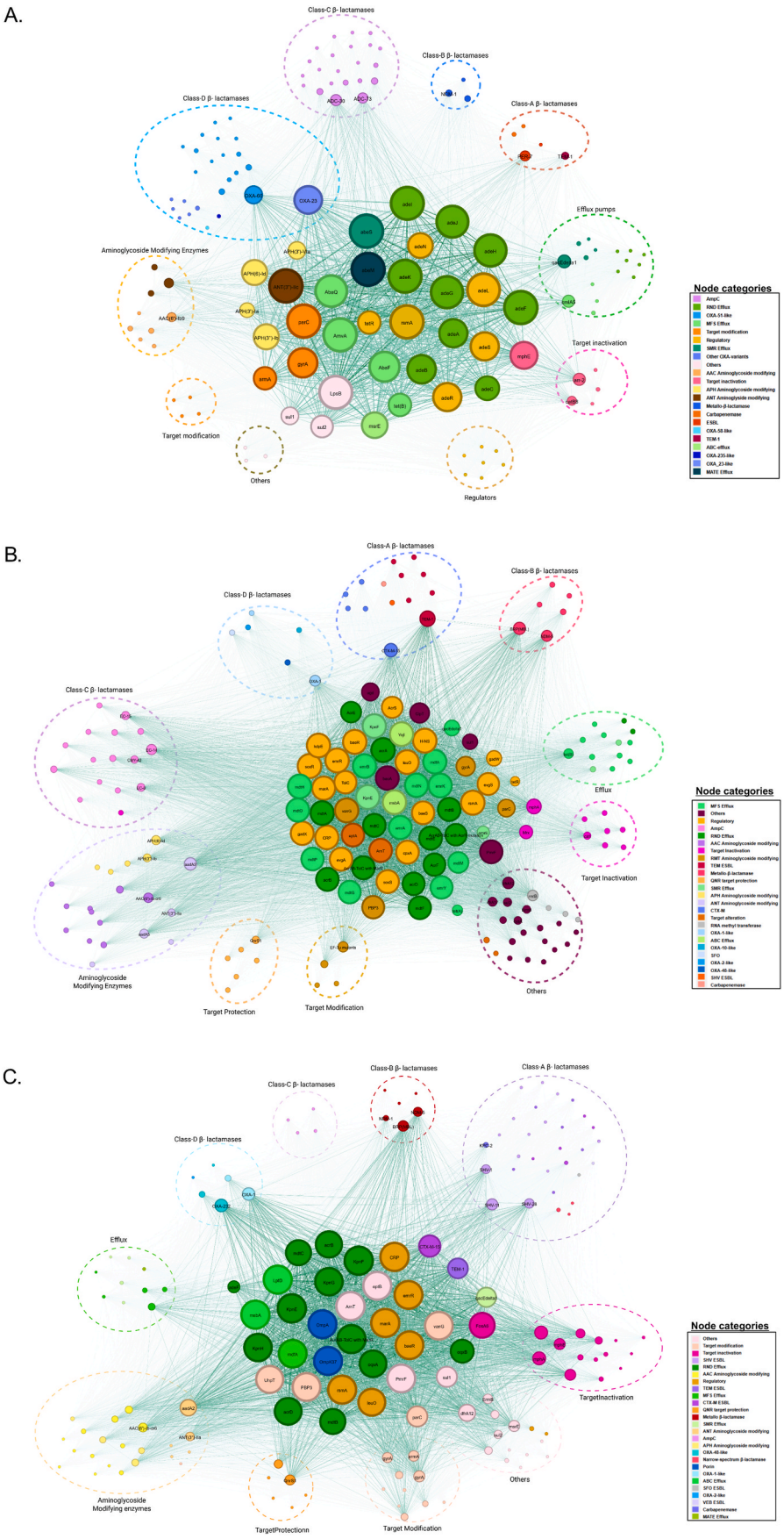


Fig. 10. Heatmap showing the presence-absence of significant ARGs ($P < 0.05$) and their association with the resistance phenotype (OR > 1, CI 95%) of Cefotaxime (A), Cefuroxime (B) in *K. pneumoniae*.
Individual ARGs are represented by the rows, and genomes are represented by columns. The heatmap shows gene presence (dark blue) and absence (light blue). Genomes are hierarchically clustered based on ARG profiles using distance-based clustering (dendrogram shown on the top). The red bar on the top shows the resistant phenotype of the genomes. The heatmap contains only those ARGs that showed a statistically significant association with the resistance phenotype of cefotaxime and cefuroxime ($P < 0.05$, Fisher's exact test).

ARGs are observed (Table S2 (Global_com)). In both the datasets, genes encoding *adeIJK* efflux system were nearly ubiquitous (95-100%), supporting previous evidence that adeIJK efflux pump is a conserved component of *A. baumannii*'s resistome [27]. Its regulator gene *adeN* was also highly widespread (India: 73.1%; Global: 86.6%). The datasets differed in the prevalence of acquired OXA carbapenemases. While the $bla_{OXA\text{-}23}$ gene was detected in over 80% of the curated genomes, only 17.2% of the global genomes harboured it. This indicates widespread dissemination of $bla_{OXA\text{-}23}$-producing *A. baumannii* in India. Intrinsic $bla_{OXA\text{-}51\text{-}like}$ variants that contribute to resistance when over-expressed were common to both groups but differed in their distribution. The global set contained a wide range of low-frequency rare bla_{OXA} variants reflecting sporadic emergence events and diverse *A. baumannii* population. Metallo-β-lactamase $bla_{NDM\text{-}1}$ was disproportionately prevalent in the Indian samples (India: 11.8%; Global: 0.6%) and $bla_{NDM\text{-}40}$ was detected solely in the curated dataset. Furthermore, PER family ESBLs were enriched in the Indian samples ($bla_{PER\text{-}7}$:23.5%; $bla_{PER\text{-}8}$:0.8%) compared to low frequencies in the global dataset ($bla_{PER\text{-}1}$: 2.8%; $bla_{PER\text{-}7}$: 0.2%), further supporting regional dissemination of specific resistance genes. Several low-frequency ARGs were detected exclusively among Indian isolates (≤2.5%, present in three or fewer isolates), including rare $bla_{OXA\text{-}51\text{-}like}$ variants ($bla_{OXA\text{-}73}$, $bla_{OXA\text{-}278}$, $bla_{OXA\text{-}337}$, $bla_{OXA\text{-}338}$, $bla_{OXA\text{-}420}$, $bla_{OXA\text{-}441}$, $bla_{OXA\text{-}927}$), $bla_{NDM\text{-}40}$, $bla_{PER\text{-}8}$, $bla_{ACT\text{-}25}$

3.4.2. *E. coli*

In the case of *E. coli*, carbapenem resistance genes found in this study (n = 305) were compared with the non-overlapping global dataset (n = 1008). The intrinsic resistance genes associated with efflux and global stress response regulation - *marA*, mutated *soxS*, and *tolC* - were detected in >90% genomes in both groups. Their universal distribution indicates that they constitute a conserved set of *E. coli*'s resistome. Comparison of the two datasets revealed a striking difference in the geographic distribution of metallo-β-lactamases. NDM enzymes were present in over 35% of Indian curated genomes, in contrast to, <1% globally. Enrichment of bla_{NDM} enzymes in the curated dataset was due to the dominance of $bla_{NDM\text{-}5}$ (Indian: 31.8%; Global: 0.3%) along with



(caption on next page)

Fig. 11. ARG co-occurrence networks of (A) *A. baumannii*, (B) *E. coli and (C) K. pneumoniae*
ARGs are represented as nodes in the networks, sized according to their prevalence in the dataset and coloured according to their functions. Edges represent co-occurrence of connected ARGs within the same isolate, and the scaled edge weights, i.e., the frequency of co-occurrence, have been represented by the intensity edge colour and thickness.

detection of other bla_{NDM} variants (bla_{NDM-1}, bla_{NDM-4}, bla_{NDM-7}, bla_{NDM-15}). $bla_{OXA-48-like}$ enzymes also showed higher frequencies in the Indian samples. For example, $bla_{OXA-181}$ was detected in 3.9%, i.e., in 12 Indian genomes and was rare in the global dataset showing presence only in a single genome. KPC enzymes, on the other hand, were more prevalent in the global dataset (bla_{KPC-2}: 1.2%; bla_{KPC-3}: 0.7%) than the Indian genomes (bla_{KPC-2} was detected in a single genome).

The same datasets were also compared for the distribution of cephalosporin-associated resistance determinants. Genes of the AcrAB-TolC system, *marA*, *soxS*, *soxR*, *kpnEF* genes, etc. that are involved in efflux pumps or regulation were present in ~95-100% isolates in both datasets suggesting that these might be universally conserved in *E. coli*. The occurrence of $bla_{CTX-M-15}$ was nearly three times higher in the Indian dataset (43.9%) than the global genomes (14.4%). bla_{OXA-1} and bla_{TEM-1}, are also more prevalent in the Indian dataset than the global set of genomes. EC-family AmpC variants bla_{EC-14}, bla_{EC-15} and bla_{EC-8} were more common in Indian samples but bla_{EC-5} is much more prevalent in the global dataset (70.1%) than in the Indian samples (3.3%). This indicates geographical variation in the distribution of EC-family *ampC* variants (Table S3 (Global_com)).

3.4.3. K. pneumoniae

Thirty distinct carbapenem resistance-associated genes were detected in the 447 curated Indian *K. pneumoniae* isolates and twenty-five distinct carbapenem determinants were detected in a global dataset of 1912 *K. pneumoniae* genomes using RGI. The two groups shared a set eleven ARGs among which the intrinsic genes *marA*, *ompK37* and the *kpnGH* pump genes were the most prevalent universally (>96% genomes). KPC carbapenemases were seen to be globally distributed, having similar frequency in the two groups, with bla_{KPC-2} being the predominant KPC gene in India. The metallo-β-lactamases, majorly bla_{NDM-5} and bla_{NDM-1}, were much higher in the Indian dataset (bla_{NDM-5}: 25.1% and bla_{NDM-1}: 11.4%) compared to their global presence (in ~5% genomes) (Table S4 (ARGs_AST)). $bla_{OXA-48-like}$ family genes were highly enriched in the Indian dataset: $bla_{OXA-232}$ was detected in 38.5% genomes in the Indian dataset and was seen in only 1.6% genomes from the global dataset. Similarly, $bla_{OXA-181}$ was present in 13% Indian isolates, as compared to, 0.4% global genomes. Similar to *A. baumannii*'s results, the global genomes presented a broader diversity of carbapenemases - VIM enzymes, rare OXA variants and multiple KPC variants, while the Indian isolates were dominated by a few highly prevalent resistance genes.

Cephalosporin-associated resistance determinants were compared in the *K. pneumoniae* datasets of 447 Indian isolates and the global dataset of 1912 genomes. *acrAB*, *kpnEF*, *marA* and *ompK3* genes were detected in high frequencies in nearly all isolates of both the groups. The globally dominant ESBL, $bla_{CTX-M-15}$, had high prevalence in both datasets (Indian: 73.4%; Global: 41.4%) but was more enriched in Indian samples. A similar distribution was seen for bla_{TEM-I} in the two groups as well. bla_{OXA-1} occurred in 38% of Indian isolates versus 28.1% global set isolates. Carbapenemase genes like bla_{NDM-5}, bla_{NDM-1} and $bla_{OXA-232}$ are also observed in higher frequencies in the Indian dataset, highlighting the emergence of highly resistant *K. pneumoniae* lineages. The global dataset contained a wide range of low-frequency SHV variants, in contrast to the Indian isolates that were dominated by a smaller set of high-prevalence β-lactamases (Table S4 (Global_com)). It is important to note that the comparisons presented here for all the three pathogens are made with quality checked genomes in ARKbase.

In this study, it is evident that the genome sequencing data along with AST is generated across several institutions. However, very limited data (genomes and AST) on Indian clinical isolates was obtained from repositories like BV-BRC and AST Browser using their search features. In addition, in literature, summaries of AST data are provided along with BioProject IDs for genomic datasets. However, the isolate level AST data is missing, limiting further investigations. These observations reflect the lack of standardized data reporting for AMR. In addition, several metadata are missing that are relevant for a complete understanding of AMR emergence and its impact, for e.g., isolation source, geographic location, isolation date along with the performed AST and genome data accessions. For all three pathogens, a total of 871 samples were analyzed, exact geographic location for 73 samples could not be confirmed despite personal communication with the authors. Similarly, the isolation source for 77 samples across all three pathogens remained unknown. In addition, it is also observed that most genomes are draft limiting the investigations to probe genomic rearrangements or HGTs. In addition, there is heterogeneity in reporting ARGs, for e.g., reporting ARG from family to variant level annotations, the databases referred to are not consistent, the alignment criteria are varied, reference genomes used are also not consistent, etc.

3.5. Recommendations towards standardised data reporting

Advances in molecular diagnostic techniques, particularly next generation sequencing, can lead to personalized and actionable insights within hours of sample acquisition. Adoption of whole genome sequencing as a preferred method in routine clinical microbiology laboratories is expected to rapidly populate public databases. In this context, standardization for data storage, analysis reports and metadata completeness will define various ways in which data may be used for various downstream applications. Standardization of metadata reporting should be made mandatory for reporting of AMR studies as it directly impacts the downstream analysis. There has also been a surge in genome sequencing across the globe and this has led to development of AI/ML methods to identify novel ARGs and other non-coding factors that can contribute to resistance phenotype. Even in the Indian context, new antibiotic resistant determinants are expected that may contribute to AMR. Therefore, in order to realize the full potential of multiomics datasets the data needs to be presented in standard metadata structure including some basic parameters (clinical, microbiological and molecular) like antibiotics exposure, dosage, treatment details, patient outcomes, other comorbid conditions, exposure to drugs other than antibiotics, site of infection, details on sequencing and data analysis protocols, etc. A centralised resource based on FAIR principles will help in data sharing and will enable removing discrepancies in data generation reporting and analysis. A standard metadata architecture will not only enable convergence of AMR datasets across clinical contexts but also facilitate comparisons with other verticals like environmental and animal AMR to understand the transmission dynamics of ARGs. This will provide insights on evolution of strains across the globe and can lead to prediction of novel drug resistance determinants in different geographical settings. Several global and national efforts are ongoing for active surveillance of AMR like GLASS and AMRSN. In addition, several comprehensive resources like CARD, NDARO, Harmonization, BV-BRC, etc. also exist that provide information on drug resistant determinants on a wide array of pathogens. However, lack of standardization across databases (conflicting gene names and synonyms), inability to represent complex resistance mechanisms (like epistatic relationships between multiple genes), unavailability of platforms for cross-comparison of AMR determinants across priority pathogens, the absence of a consensus reference genome (more than one reference genomes), etc. are several

open challenges. Some of these challenges are addressed by newly developed resources like ARKbase [24] and AMRNet [45]. To further strengthen the applications of these resources, it is imperative to integrate extensive data analysis workflows in these platforms [46]. For example, there are several machine learning models available for understanding AMR but it is hard to determine their utility in specific contexts, as they are not readily available for execution [47]. With the availability of the proposed platform, the scientific community working in the field of AMR will be able to use existing models on their datasets & contribute new datasets to the platform. Similarly, clinicians will be able to perform drug resistance profiling without getting into the hassles of installing and doing data preprocessing, which in most cases need expert/trained human resources. Therefore, the platform is expected to overcome challenges like standardization of protocols for genome data quality check, assembly & annotation of AMR genes. This is important as different bioinformatics pipelines lead to different outcomes. A common interface will allow open sharing of these discrepancies and will also lead to the development of a better understanding of the correlation of AST by WGS across pathogen-antibiotic combinations.

4. Conclusions

To the best of our knowledge, this is the first study focusing on systematic curation and analysis of WHO priority pathogens in the Indian context. Unlike existing studies that are typically limited to a single organism, narrow geographic coverage, or specific resistance mechanisms, this work generates a pan-India landscape for the three critical pathogens following a standardized pipeline on all the curated genomes. Through this study, variant-level resolution of ARGs across Indian isolates is enabled. The analysis also identified significant differences in the distribution of ARGs in resistant versus susceptible isolates with statistical tests towards genotype-phenotype associations. Overall, this study advances beyond fragmented literature by providing a comprehensive and standardized view of AMR in the Indian landscape, allowing better identification of ARG patterns, enriched and globally distinct, in Indian isolates.

Funding

The Indian Council of Medical Research (grant/award number: 'IIRP-2023-2784'), Council of Scientific and Industrial Research (India) (grant/award number: 'HCP-047'), Funding to pay the Open Access publication charges for this article was provided by the Council of Scientific and Industrial Research (India) ('OLP0213-20').

CRediT authorship contribution statement

Rupali Aggarwal: Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Ankita Das:** Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Jasleen Kaur:** Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Anshu Bhardwaj:** Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

I have nothing to declare.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bbrc.2026.153871.

References

[1] Antimicrobial resistance: a silent pandemic, Nat. Commun. 15 (1) (2024 Jul 23) 6198, https://doi.org/10.1038/s41467-024-50457-z. PubMed PMID: 39043632; PubMed Central PMCID: PMC11266683.
[2] M. Naghavi, S.E. Vollset, K.S. Ikuta, L.R. Swetschinski, A.P. Gray, E.E. Wool, et al., Global burden of bacterial antimicrobial resistance 1990–2021: a systematic analysis with forecasts to 2050, Lancet 404 (10459) (2024 Sep 28) 1199–1226, https://doi.org/10.1016/S0140-6736(24)01867-1.
[3] H. Sati, E. Carrara, A. Savoldi, P. Hansen, J. Garlasco, E. Campagnaro, et al., The WHO bacterial priority pathogens list 2024: a prioritisation study to guide research, development, and public health strategies against antimicrobial resistance, Lancet Infect. Dis. 25 (9) (2025 Sep 1) 1033–1043, https://doi.org/10.1016/S1473-3099(25)00118-5. PubMed PMID: 40245910.
[4] Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis, Lancet Lond. Engl. 399 (10325) (2022 Feb 12) 629–655, https://doi.org/10.1016/S0140-6736(21)02724-0. PubMed PMID: 35065702; PubMed Central PMCID: PMC8841637.
[5] C.A.D. Burnham, J. Leeds, P. Nordmann, J. O'Grady, J. Patel, Diagnosing antimicrobial resistance, Nat. Rev. Microbiol. 15 (11) (2017 Nov) 697–703, https://doi.org/10.1038/nrmicro.2017.103.
[6] M. Boolchandani, A.W. D'Souza, G. Dantas, Sequencing-based methods and resources to study antimicrobial resistance, Nat. Rev. Genet. 20 (6) (2019 Jun) 356–370, https://doi.org/10.1038/s41576-019-0108-4. PubMed PMID: 30886350; PubMed Central PMCID: PMC6525649.
[7] F.O. Bagger, L. Borgwardt, A.S. Jespersen, A.R. Hansen, B. Bertelsen, M. Kodama, et al., Whole genome sequencing in clinical practice, BMC Med. Genom. 17 (2024 Jan 29) 39, https://doi.org/10.1186/s12920-024-01795-w. PubMed PMID: 38287327; PubMed Central PMCID: PMC10823711.
[8] A. Tsakris, A. Ikonomidis, S. Pournaras, N. Spanakis, A. Markogiannakis, Carriage of OXA-58 but not of OXA-51 β-lactamase gene correlates with carbapenem resistance in Acinetobacter baumannii, J. Antimicrob. Chemother. 58 (5) (2006 Nov 1) 1097–1099, https://doi.org/10.1093/jac/dkl365.
[9] S. Rangama, I.D.E.A. Lidbury, J.M. Holden, C. Borsetto, A.R.J. Murphy, P. M. Hawkey, et al., Mechanisms involved in the active secretion of CTX-M-15 β-Lactamase by pathogenic Escherichia coli ST131, Antimicrob. Agents Chemother. 65 (10) (2021 Sep 17) e0066321, https://doi.org/10.1128/AAC.00663-21. PubMed PMID: 34310213; PubMed Central PMCID: PMC8448145.
[10] E. Avershina, P. Sharma, A.M. Taxt, H. Singh, S.A. Frye, K. Paul, et al., AMR-Diag: Neural network based genotype-to-phenotype prediction of resistance towards β-lactams in Escherichia coli and Klebsiella pneumoniae, Comput. Struct. Biotechnol. J. 19 (2021) 1896–1906, https://doi.org/10.1016/j.csbj.2021.03.027. PubMed PMID: 33897984; PubMed Central PMCID: PMC8060595.
[11] C.S. Ho, C.T.H. Wong, T.T. Aung, R. Lakshminarayanan, J.S. Mehta, S. Rauz, et al., Antimicrobial resistance: a concise update, Lancet Microbe 6 (1) (2025 Jan 1), https://doi.org/10.1016/j.lanmic.2024.07.010. PubMed PMID: 39305919.
[12] R.D. Olson, R. Assaf, T. Brettin, N. Conrad, C. Cucinell, J.J. Davis, et al., Introducing the bacterial and viral bioinformatics resource center (BV-BRC): a resource combining PATRIC, IRD and ViPR, Nucleic Acids Res. 51 (D1) (2023 Jan 6) D678–D689, https://doi.org/10.1093/nar/gkac1003. PubMed PMID: 36350631; PubMed Central PMCID: PMC9825582.
[13] D. Yamin, V. Uskoković, A.M. Wakil, M.D. Goni, S.H. Shamsuddin, F.H. Mustafa, et al., Current and future technologies for the detection of antibiotic-resistant bacteria, Diagn. Basel Switz. 13 (20) (2023 Oct 18), https://doi.org/10.3390/diagnostics13203246. PubMed PMID: 37892067; PubMed Central PMCID: PMC10606640.
[14] M.J. Ellington, O. Ekelund, F.M. Aarestrup, R. Canton, M. Doumith, C. Giske, et al., The role of whole genome sequencing in antimicrobial susceptibility testing of bacteria: report from the EUCAST subcommittee, Clin. Microbiol. Infect. Off. Publ. Eur. Soc. Clin. Microbiol. Infect. Dis. 23 (1) (2017 Jan) 2–22, https://doi.org/10.1016/j.cmi.2016.11.012. PubMed PMID: 27890457.
[15] Annual Report National Antimicrobial Resistance Surveillance Network (NARS-Net), 2024. Available from: https://ncdc.mohfw.gov.in/uploads/pdf/amr30.pdf. (Accessed 13 February 2026).
[16] S. Andrews, FastQC: a Quality Control Tool for High Throughput Sequence Data, 2010. https://www.bioinformatics.babraham.ac.uk/projects/fastqc/.
[17] S. Chen, Y. Zhou, Y. Chen, J. Gu, Fastp: an ultra-fast all-in-one FASTQ preprocessor, Bioinformatics 34 (17) (2018 Sep 1) i884–i890, https://doi.org/10.1093/bioinformatics/bty560.
[18] A. Bankevich, S. Nurk, D. Antipov, A.A. Gurevich, M. Dvorkin, A.S. Kulikov, et al., SPAdes: a new genome assembly algorithm and its applications to single-cell sequencing, J. Comput. Biol. 19 (5) (2012 May) 455–477, https://doi.org/10.1089/cmb.2012.0021. PubMed PMID: 22506599; PubMed Central PMCID: PMC3342519.
[19] A. Gurevich, V. Saveliev, N. Vyahhi, G. Tesler, QUAST: quality assessment tool for genome assemblies, Bioinfor. Oxf. Engl. 29 (8) (2013 Apr 15) 1072–1075, https://doi.org/10.1093/bioinformatics/btt086. PubMed PMID: 23422339; PubMed Central PMCID: PMC3624806.
[20] B.P. Alcock, W. Huynh, R. Chalil, K.W. Smith, A.R. Raphenya, M.A. Wlodarski, et al., CARD 2023: expanded curation, support for machine learning, and resistome prediction at the comprehensive antibiotic resistance database, Nucleic Acids Res. 51 (D1) (2023 Jan 6) D690–D699, https://doi.org/10.1093/nar/gkac920. PubMed PMID: 36263822; PubMed Central PMCID: PMC9825576.
[21] Z. Xie, H. Tang, ISEScan: automated identification of insertion sequence elements in prokaryotic genomes, Bioinformatics 33 (21) (2017 Nov 1) 3340–3347, https://doi.org/10.1093/bioinformatics/btx433. PubMed PMID: 29077810.

[22] J. Robertson, J.H.E. Nash, MOB-suite: software tools for clustering, reconstruction and typing of plasmids from draft assemblies, Microb. Genom. 4 (8) (2018 Jul 27) e000206, https://doi.org/10.1099/mgen.0.000206. PubMed PMID: 30052170; PubMed Central PMCID: PMC6159552.
[23] M. Bastian, S. Heymann, M. Jacomy, Gephi: an open source software for exploring and manipulating networks, Proc. Int. AAAI Conf. Web Soc. Media 3 (1) (2009 Mar 19) 361–362, https://doi.org/10.1609/icwsm.v3i1.13937.
[24] S. Gambhir, S. Pandey, H. Bajetha, J. Kaur, A. Das, G. Pranavathiyani, et al., ARKbase: antimicrobial resistance Knowledgebase1.0, Nucleic Acids Res. 22 (2025 Dec), https://doi.org/10.1093/nar/gkaf1307. PubMed PMID: 41428500.
[25] K. Gupta, T.M. Hooton, K.G. Naber, B. Wullt, R. Colgan, L.G. Miller, et al., International clinical practice guidelines for the treatment of acute uncomplicated cystitis and pyelonephritis in women: a 2010 update by the infectious diseases society of America and the european society for microbiology and infectious diseases, Clin. Infect. Dis. Off. Publ. Infect. Dis. Soc. Am. 52 (5) (2011 Mar 1) e103–e120, https://doi.org/10.1093/cid/ciq257. PubMed PMID: 21292654.
[26] Borde K, Dass M, Sharma RM, Mathai D. Antibiotic susceptibility testing (AST) practices in India: need for a national AST forum. Cureus. 14(11):e30971. doi: 10.7759/cureus.30971 PubMed PMID: 36465196; PubMed Central PMCID: PMC9714383.
[27] G. Kahlmeter, C.G. Giske, T.J. Kirn, S.E. Sharp, Point-counterpoint: differences between the european committee on antimicrobial susceptibility testing and clinical and laboratory standards institute recommendations for reporting antimicrobial susceptibility results, J. Clin. Microbiol. 57 (9) (2019 Aug 26), https://doi.org/10.1128/jcm.01129-19 doi:10.1128/jcm.01129-19.
[28] J.F. Turton, N. Woodford, J. Glover, S. Yarde, M.E. Kaufmann, T.L. Pitt, Identification of Acinetobacter baumannii by detection of the blaOXA-51-like carbapenemase gene intrinsic to this species, J. Clin. Microbiol. 44 (8) (2006 Aug) 2974–2976, https://doi.org/10.1128/JCM.01021-06. PubMed PMID: 16891520; PubMed Central PMCID: PMC1594603.
[29] B. Kitchel, J.K. Rasheed, A. Endimiani, A.M. Hujer, K.F. Anderson, R.A. Bonomo, et al., Genetic factors associated with elevated carbapenem resistance in KPC-producing Klebsiella pneumoniae, Antimicrob. Agents Chemother. 54 (10) (2010 Oct) 4201–4207, https://doi.org/10.1128/AAC.00008-10. PubMed PMID: 20660684; PubMed Central PMCID: PMC2944623.
[30] Z. Hashemizadeh, G. Hatam, J. Fathi, F. Aminazadeh, H. Hosseini-Nave, M. Hadadi, et al., The spread of insertion sequences element and transposons in carbapenem resistant Acinetobacter baumannii in a hospital setting in Southwestern Iran, Infect Chemother 54 (2) (2022 Jun) 275–286, https://doi.org/10.3947/ic.2022.0022. PubMed PMID: 35706082; PubMed Central PMCID: PMC9259918.
[31] S. Vijayakumar, S. Anandan, D.P. Ms, K. Kanthan, S. Vijayabaskar, A. Kapil, et al., Insertion sequences and sequence types profile of clinical isolates of carbapenem-resistant *A. baumannii* collected across India over four year period, J. Infect. Public Health 13 (7) (2020 Jul 1) 1022–1028, https://doi.org/10.1016/j.jiph.2019.11.018.
[32] R.L. Wimalasekara, D. White, A. Kumar, Targeting Acinetobacter baumannii resistance-nodulation-division efflux pump transcriptional regulators to combat antimicrobial resistance, Npj Antimicrob. Resist. 3 (1) (2025 Jan 25) 4, https://doi.org/10.1038/s44259-024-00074-z. PubMed PMID: 39863717; PubMed Central PMCID: PMC11762787.
[33] N. Rosenfeld, C. Bouchier, P. Courvalin, B. Périchon, Expression of the resistance-nodulation-cell division pump AdeIJK in Acinetobacter baumannii is regulated by AdeN, a TetR-Type regulator, Antimicrob. Agents Chemother. 56 (5) (2012 Apr 12) 2504–2510, https://doi.org/10.1128/aac.06422-11.
[34] L. Damier-Piolle, S. Magnet, S. Brémont, T. Lambert, P. Courvalin, AdeIJK, a resistance-nodulation-cell division pump effluxing multiple antibiotics in Acinetobacter baumannii, Antimicrob. Agents Chemother. 52 (2) (2008 Feb) 557–562, https://doi.org/10.1128/AAC.00732-07. PubMed PMID: 18086852; PubMed Central PMCID: PMC2224764.
[35] F.F. Tuon, C.H. Yamada, A.P. de Andrade, L.N.V.S. Arend, D. Dos Santos Oliveira, J.P. Telles, Oral doxycycline to carbapenem-resistant Acinetobacter baumannii infection as a polymyxin-sparing strategy: results from a retrospective cohort, Braz. J. Microbiol. Publ. Braz. Soc. Microbiol. 54 (3) (2023 Sep) 1795–1802, https://doi.org/10.1007/s42770-023-01015-0. PubMed PMID: 37278889; PubMed Central PMCID: PMC10243254.
[36] Z. Wang, H. Li, J. Zhang, H. Wang, Co-Occurrence of bla(OXA-23) in the chromosome and plasmid: increased fitness in carbapenem-resistant Acinetobacter baumannii, Antibiot. Basel Switz. 10 (10) (2021 Oct 1), https://doi.org/10.3390/antibiotics10101196. PubMed PMID: 34680777; PubMed Central PMCID: PMC8532878.
[37] G. Peirano, A. Endimiani, J.D.D. Pitout, CTX-M-Producing escherichia coli: history, molecular epidemiology and laboratory detection, Infect. Drug Resist. 18 (2025 Dec 10) 6549–6560, https://doi.org/10.2147/IDR.S553853. PubMed PMID: 41399812; PubMed Central PMCID: PMC12702286.
[38] S.F. Tian, Y.Z. Chu, B.Y. Chen, H. Nian, H. Shang, ISEcp1 element in association with bla(CTX-M) genes of E. coli that produce extended-spectrum β-lactamase among the elderly in community settings, Enferm. Infecc. Microbiol. Clín. 29 (10) (2011 Dec), https://doi.org/10.1016/j.eimc.2011.07.011. PubMed PMID: 22019175.
[39] K.E. Khalid, Dissemination of class A cephalosporinases and class D carbapenemases in Escherichia coli isolates from a tertiary hospital in Sudan, Cureus 15 (8) (2023 Aug) e44365, https://doi.org/10.7759/cureus.44365. PubMed PMID: 37779806; PubMed Central PMCID: PMC10540240.
[40] B.A. Evans, S.G.B. Amyes, OXA β-lactamases, Clin. Microbiol. Rev. 27 (2) (2014 Apr) 241–263, https://doi.org/10.1128/CMR.00117-13. PubMed PMID: 24696435; PubMed Central PMCID: PMC3993105.
[41] N. Gheewalla, V. Karthikeyan, Y. Jadhav, K. Kulkarni, A. Tyagi, J. Jagannadham, et al., Genomic landscape of antimicrobial resistance in India: findings from a multi-species surveillance study, Npj Antimicrob. Resist. 4 (1) (2026 Feb 16) 13, https://doi.org/10.1038/s44259-026-00185-9. PubMed PMID: 41699191; PubMed Central PMCID: PMC12910060.
[42] Annual Report Antimicrobial Resistance Research and Surveillance Network, 2024. https://www.icmr.gov.in/icmrobject/uploads/Report/1763981012_icmramrsnannualreport2024.pdf. (Accessed 1 April 2026).
[43] M. Rozwandowicz, M.S.M. Brouwer, J. Fischer, J.A. Wagenaar, B. Gonzalez-Zorn, B. Guerra, et al., Plasmids carrying antimicrobial resistance genes in enterobacteriaceae, J. Antimicrob. Chemother. 73 (5) (2018 May 1) 1121–1137, https://doi.org/10.1093/jac/dkx488. PubMed PMID: 29370371.
[44] S. Prasad, B. Dratvova, A. Gryndlerova, M. Brajerova, P. Kabelikova, J. Tkadlec, et al., IncL plasmid-mediated dissemination of OXA-48 β-lactamase and bla (CTX-M-15) gene amplification identified via long-read sequencing in carbapenem-resistant Enterobacterales, JAC-Antimicrob. Resist. 8 (1) (2026 Feb), https://doi.org/10.1093/jacamr/dlaf254. PubMed PMID: 41510202; PubMed Central PMCID: PMC12776359.
[45] L.T. Cerdeira, Z.A. Dyson, V. Sharma, M. Maranga, E. Foster-Nyarko, M.E. Carey, et al., AMRnet: a data visualization platform to interactively explore pathogen variants and antimicrobial resistance, Nucleic Acids Res. 54 (D1) (2026 Jan 6) D691–D702, https://doi.org/10.1093/nar/gkaf1101. PubMed PMID: 41206476; PubMed Central PMCID: PMC12807673.
[46] T. Sharma, R. Kumar, J.S. Kalra, S. Singh, G.S. Bhalla, A. Bhardwaj, Galaxy ASIST: a web-based platform for mapping and assessment of global standards of antimicrobial susceptibility: a case study in Acinetobacter baumannii genomes, Front. Microbiol. (2023 Jan 27) 13, https://doi.org/10.3389/fmicb.2022.1041847.
[47] A.L. Hicks, N. Wheeler, L. Sánchez-Busó, J.L. Rakeman, S.R. Harris, Y.H. Grad, Evaluation of parameters affecting performance and reliability of machine learning-based antibiotic susceptibility testing from whole genome sequencing data, PLoS Comput. Biol. 15 (9) (2019 Sep) e1007349, https://doi.org/10.1371/journal.pcbi.1007349. PubMed PMID: 31479500; PubMed Central PMCID: PMC6743791.